Florida Keys National Marine Sanctuary

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

Contents

Indicators

Nutrients

Total Nitrogen - Discrete

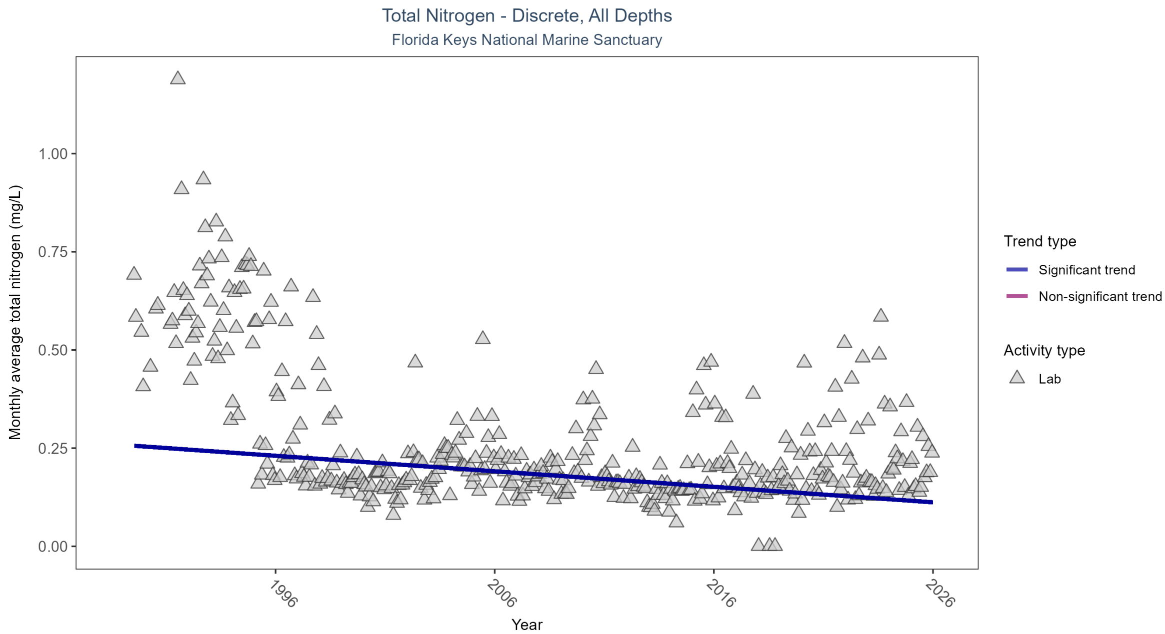

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen decreased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	36432	37	1989 - 2025	0.145	-0.27923	0.25822	-0.00395	0

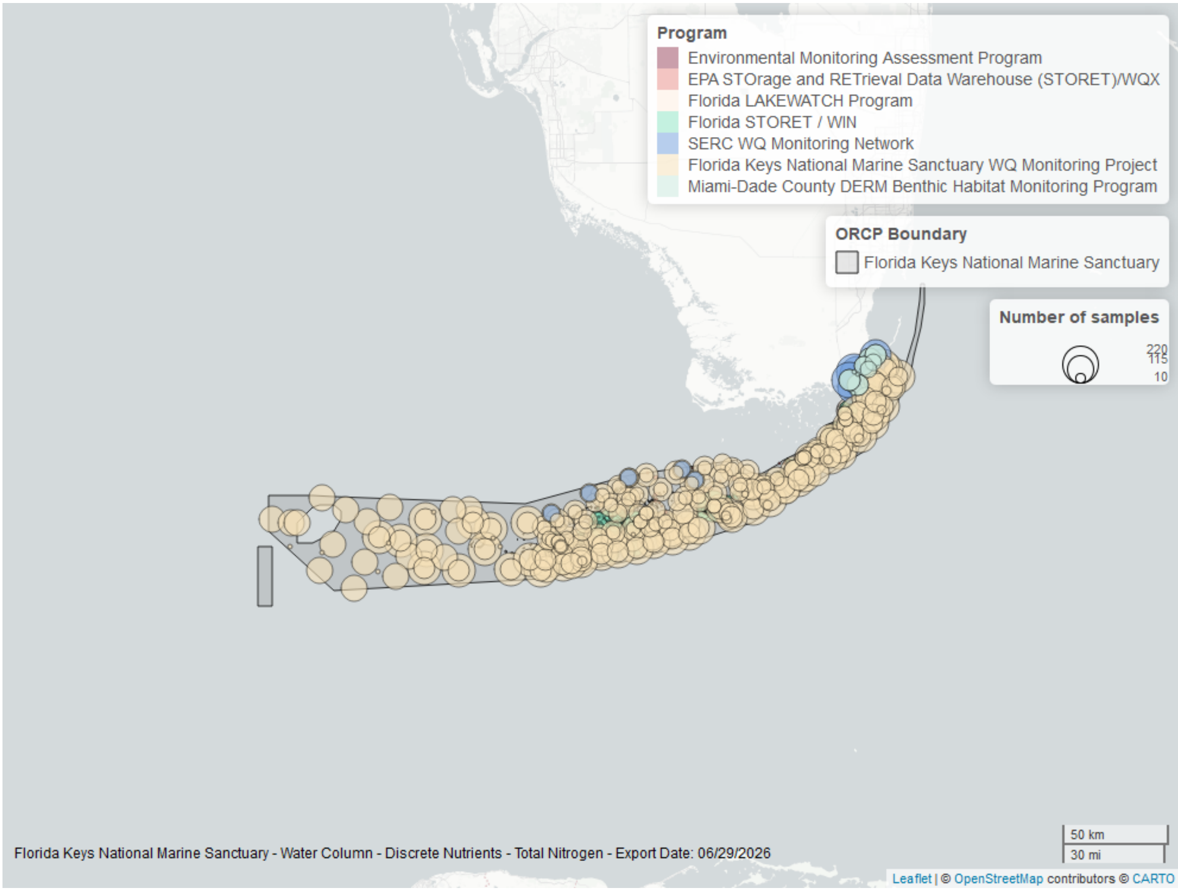

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

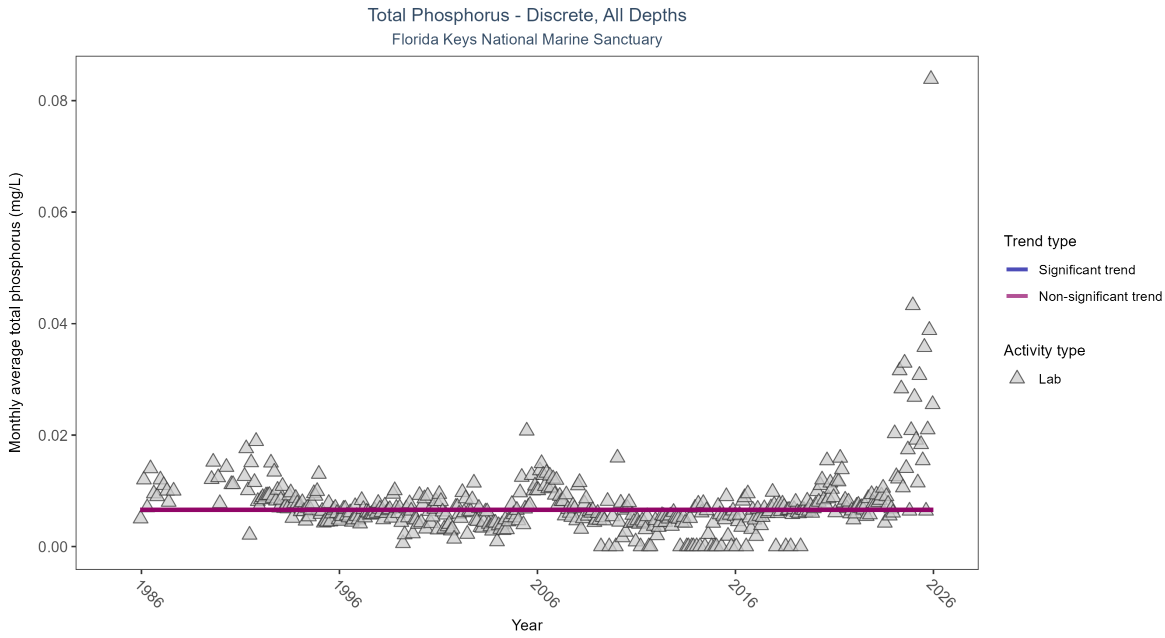

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1985 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	34316	40	1985 - 2025	0.006	-0.00212	0.0066	0	0.94

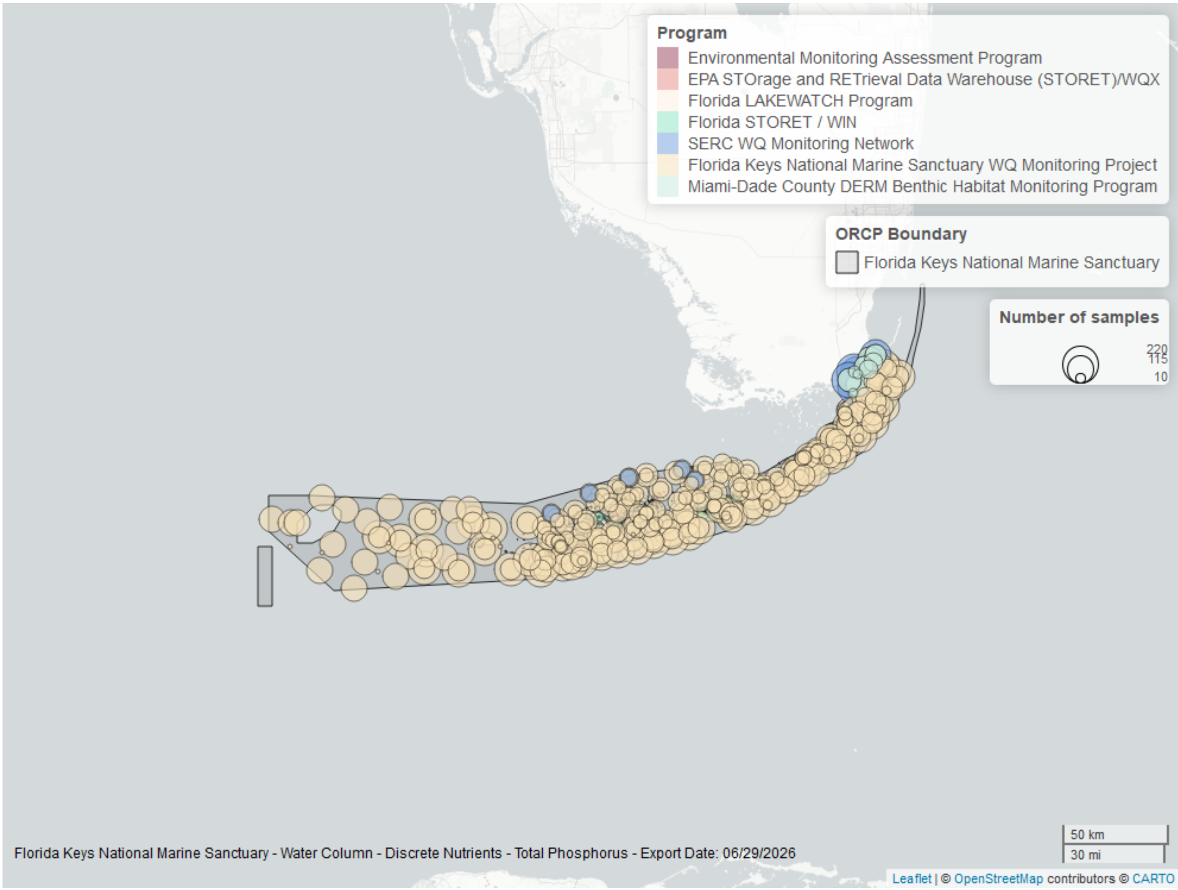

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

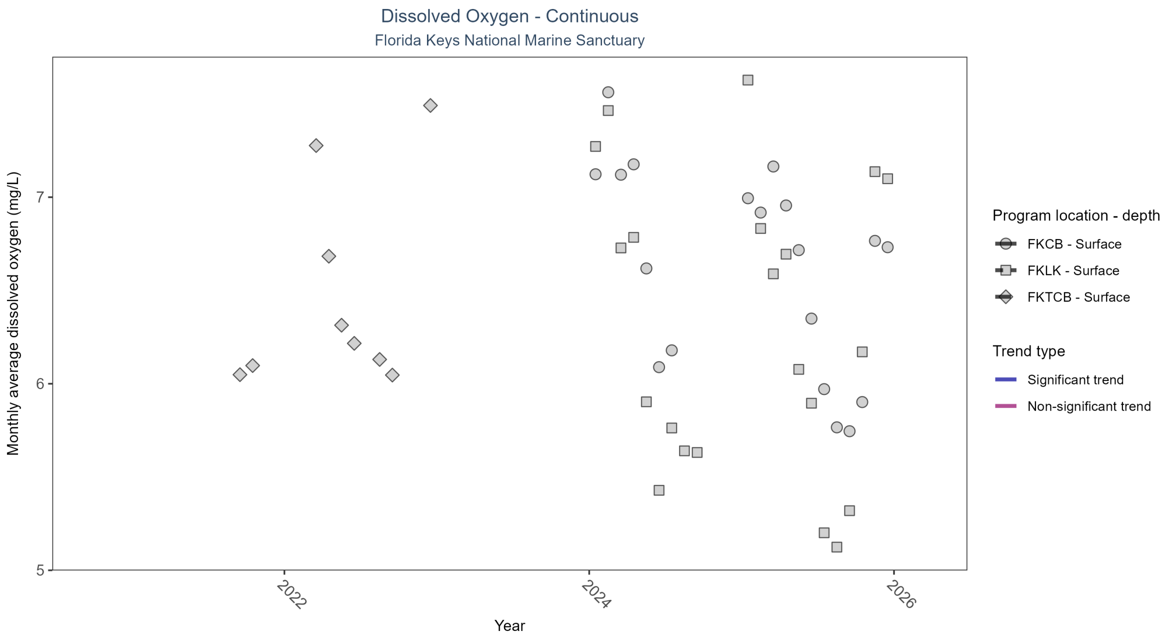

Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	46023	2	2024 - 2025	6.5	-	-	-	-
FKLK	Insufficient data	51715	2	2024 - 2025	6.2	-	-	-	-
FKTCB	Insufficient data	13514	2	2021 - 2022	6.4	-	-	-	-

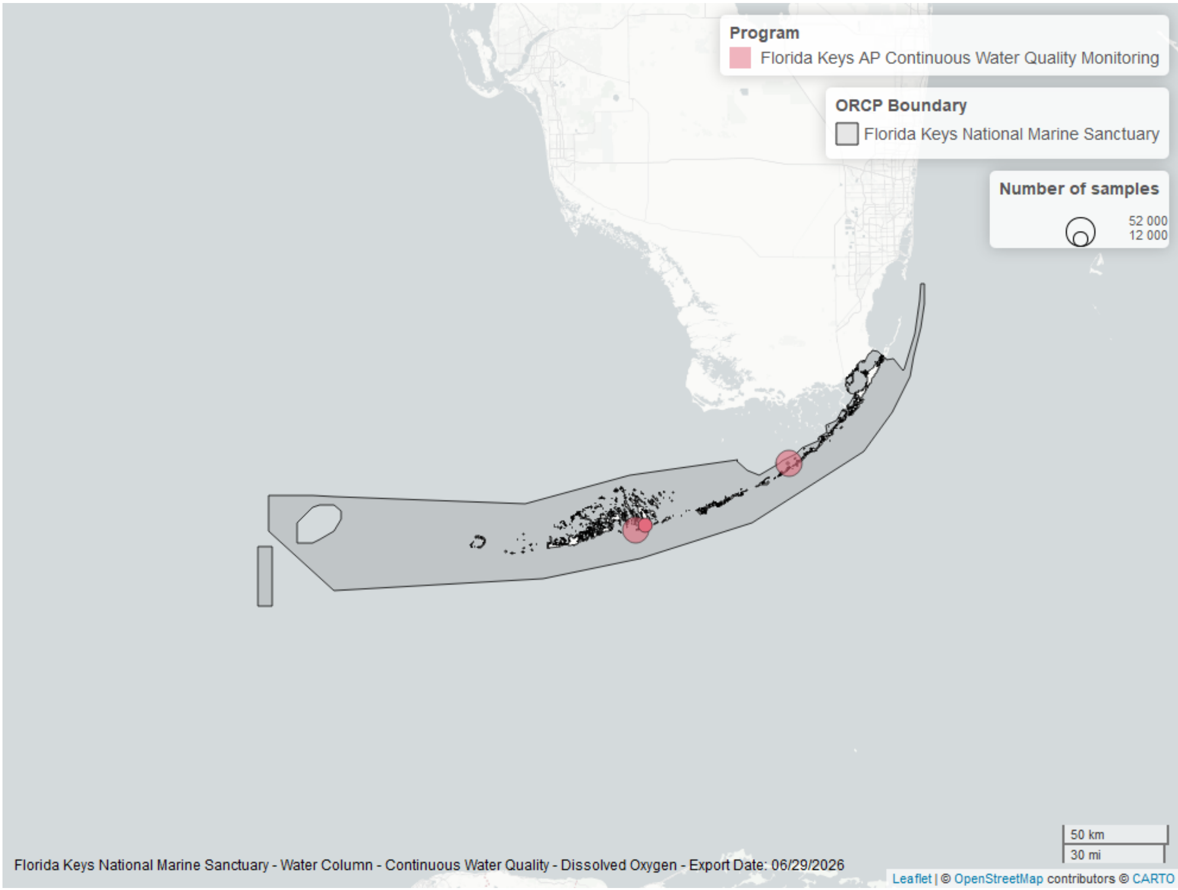

Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

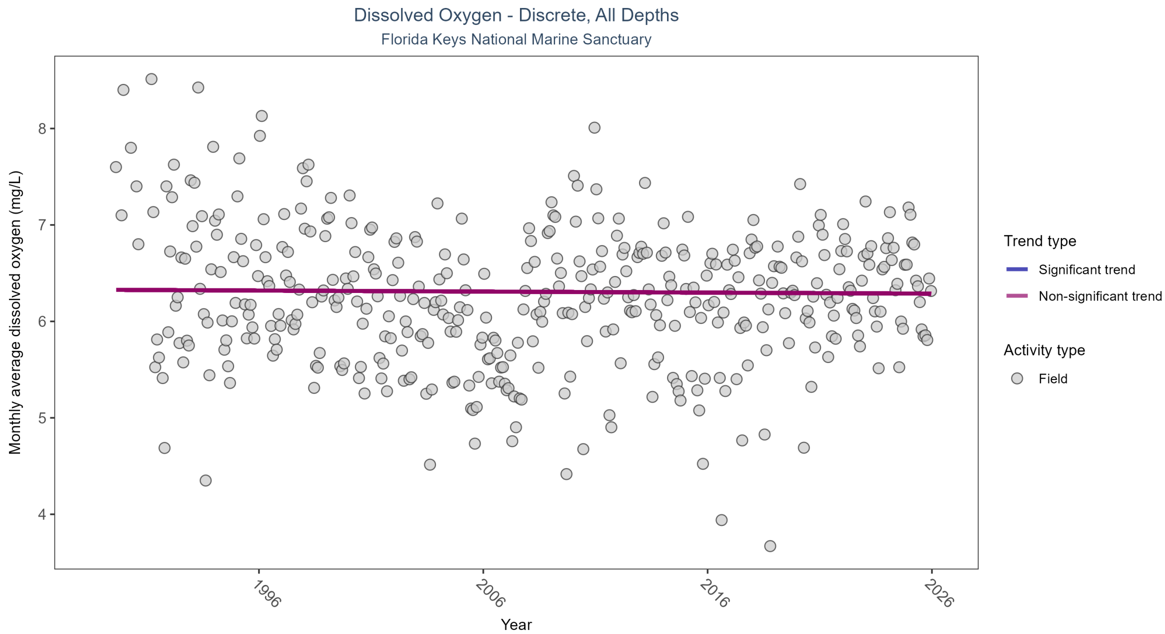

Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Dissolved oxygen showed no detectable trend between 1989 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	54598	37	1989 - 2025	6.3	-0.01464	6.3276	-0.00108	0.6653

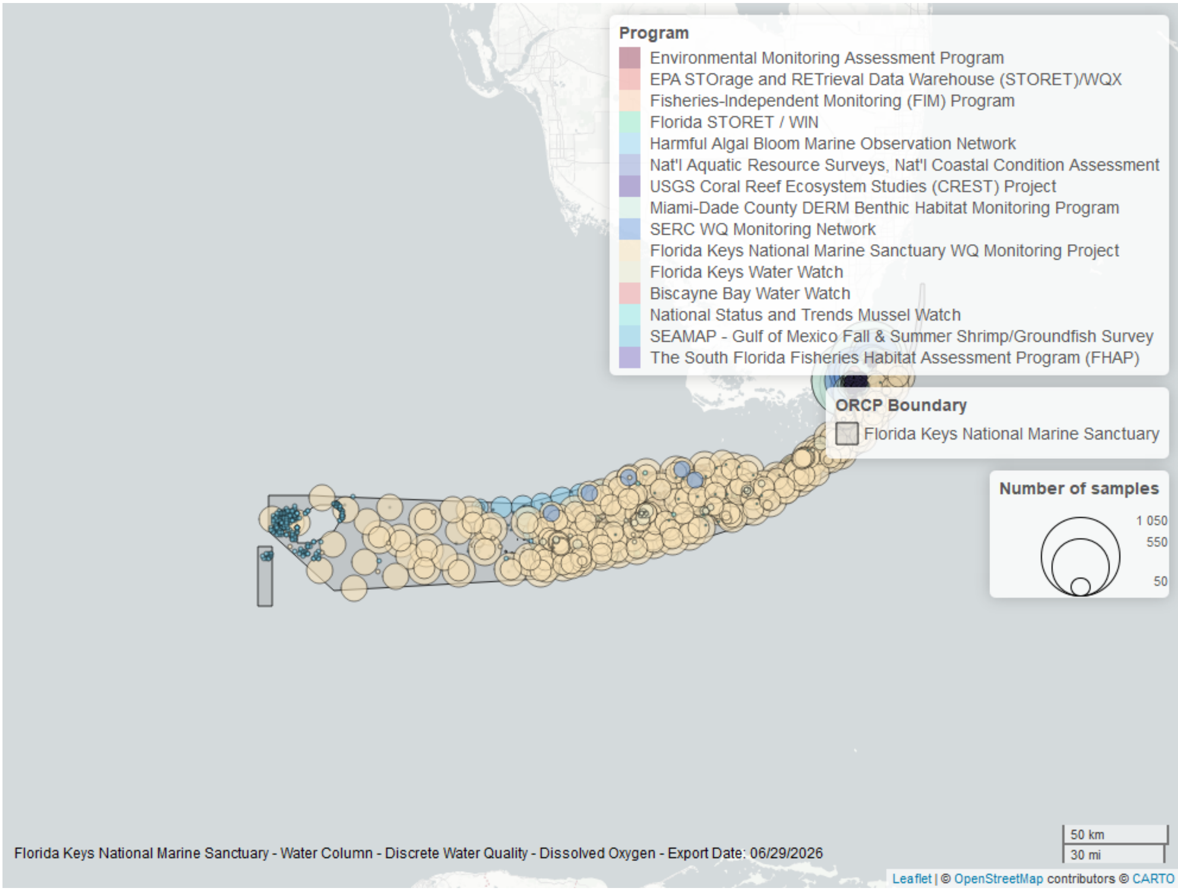

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

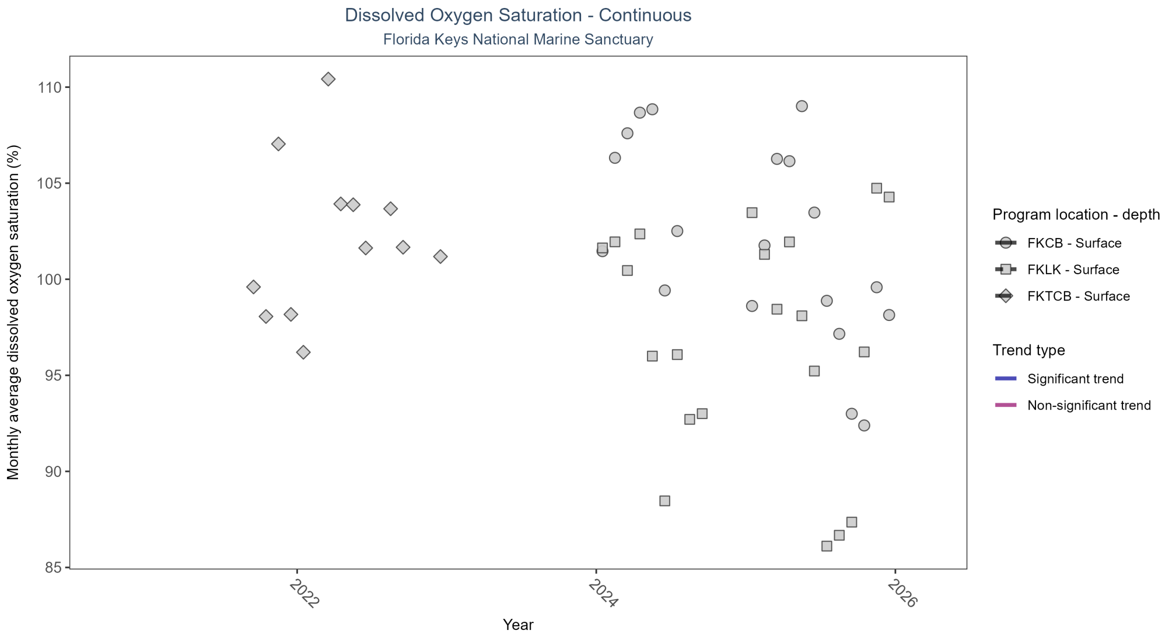

Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	46055	2	2024 - 2025	100.5	-	-	-	-
FKLK	Insufficient data	51731	2	2024 - 2025	93.6	-	-	-	-
FKTCB	Insufficient data	17063	2	2021 - 2022	100.2	-	-	-	-

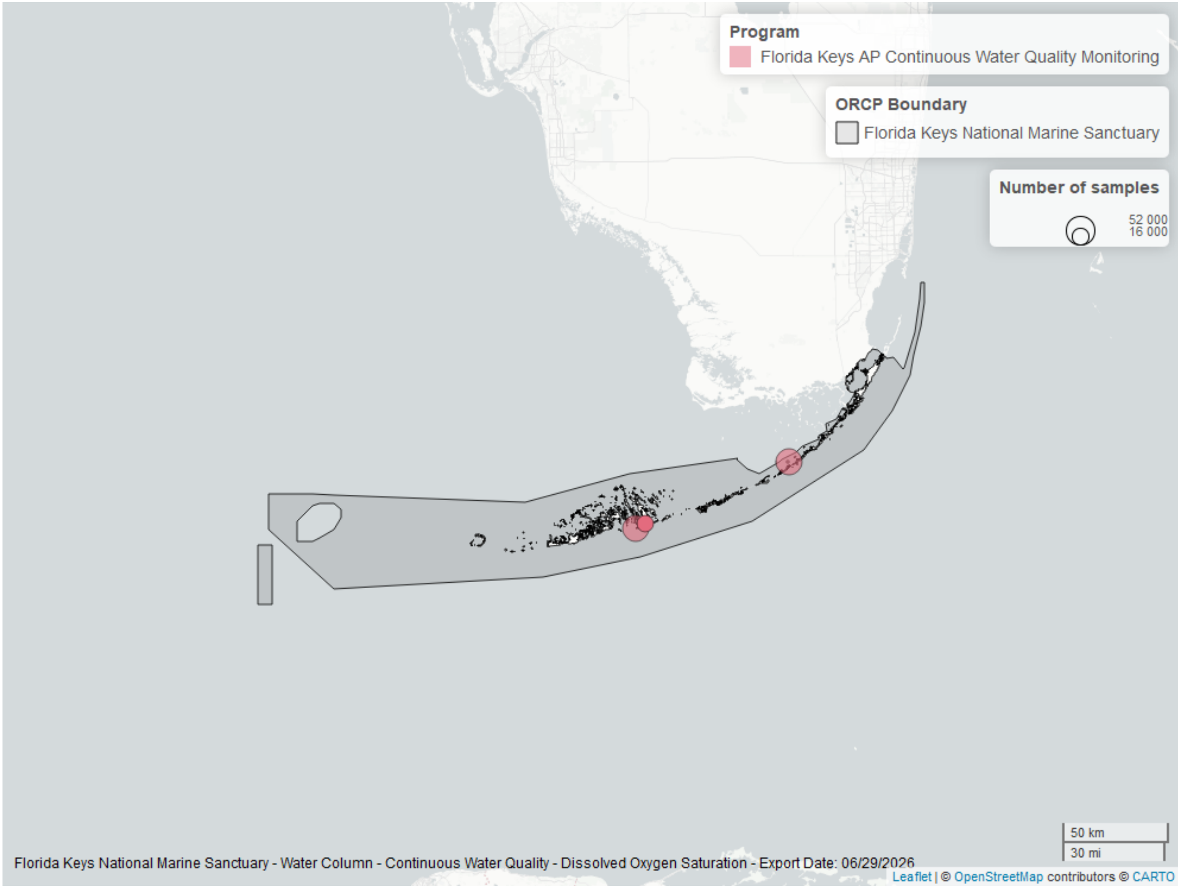

Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

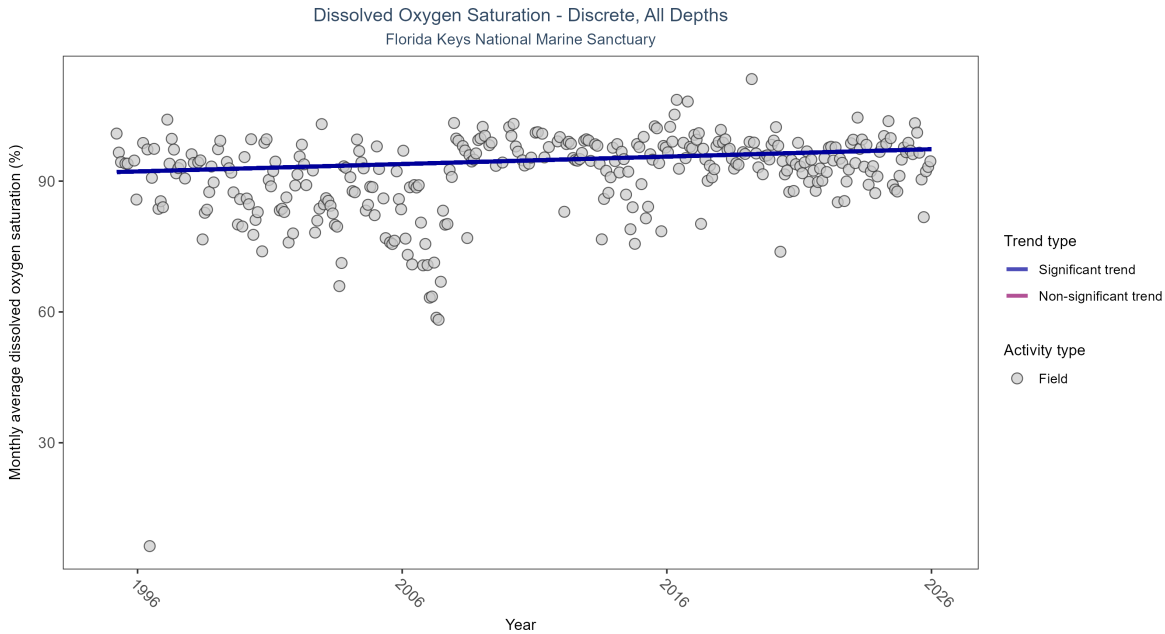

Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation increased by 0.17% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	30582	31	1995 - 2025	94.73586	0.17712	92.04785	0.17005	0

\end{table}

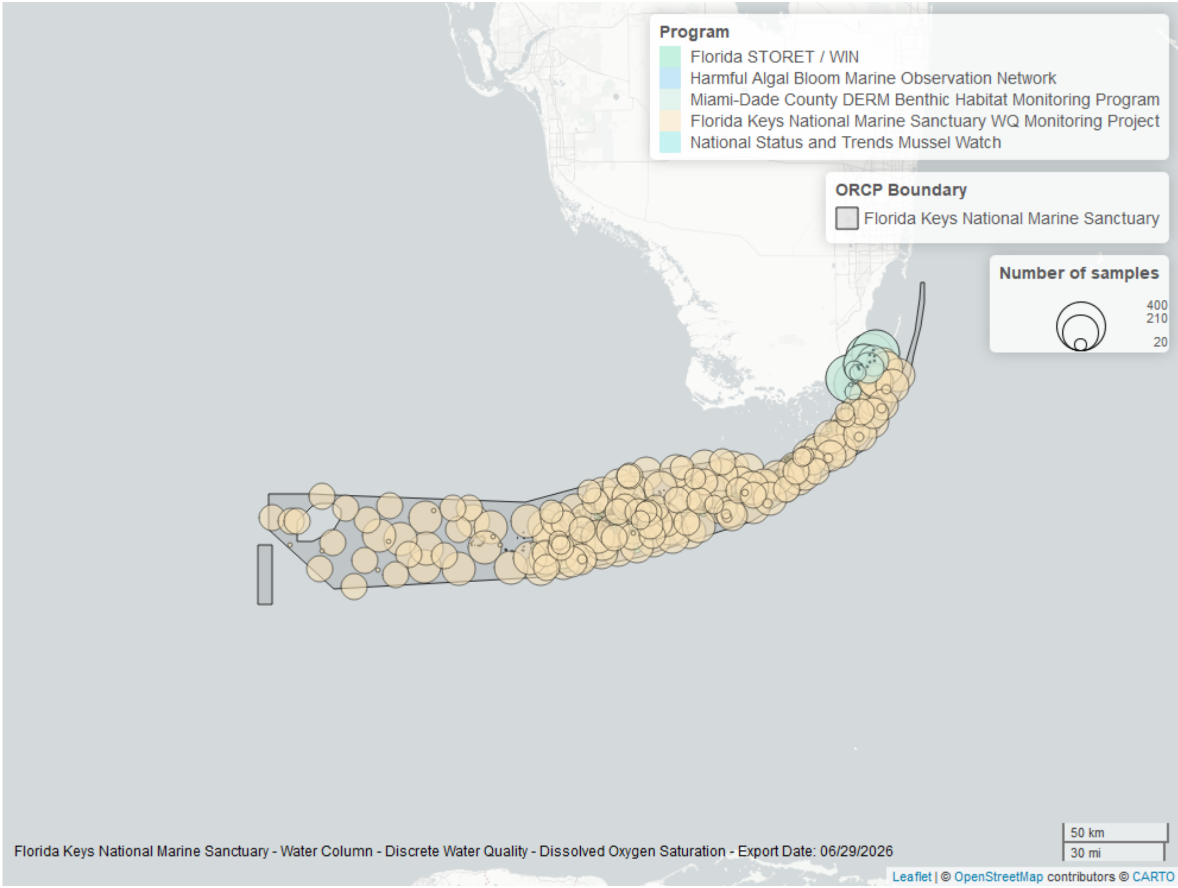

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

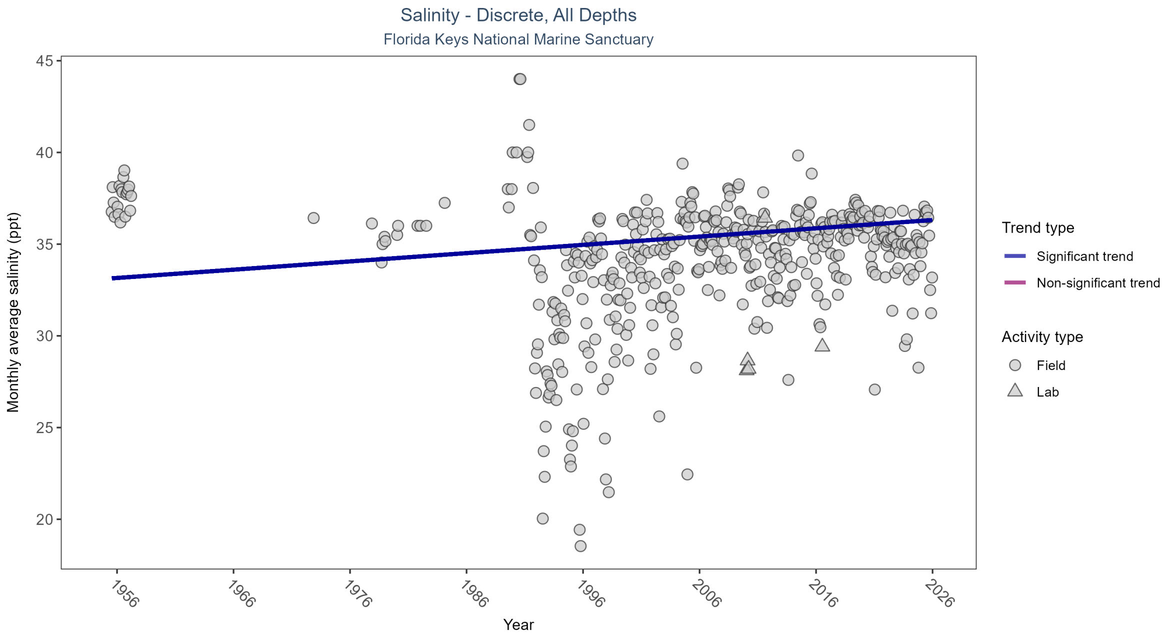

Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 6: Monthly average salinity increased by 0.05 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	62689	48	1955 - 2025	36.0929	0.13255	33.11127	0.04512	0.0001

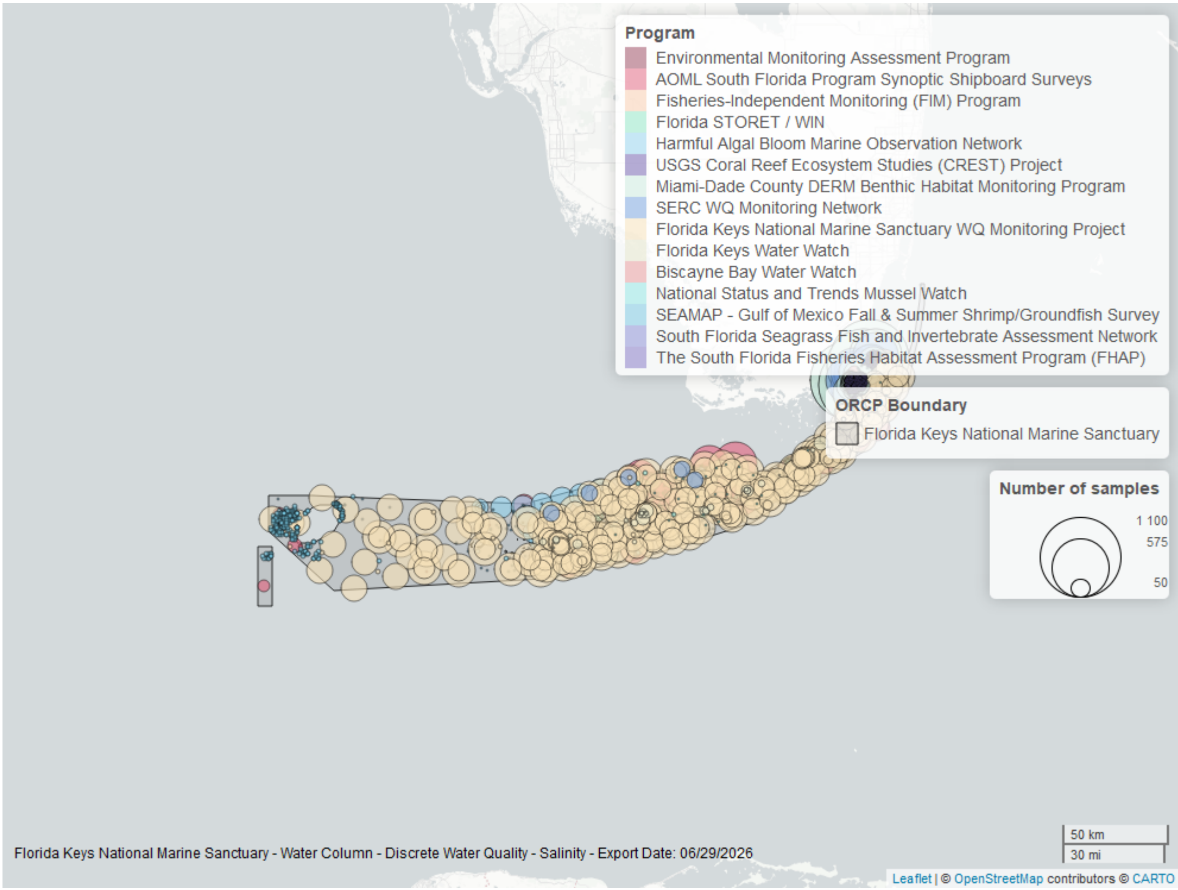

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

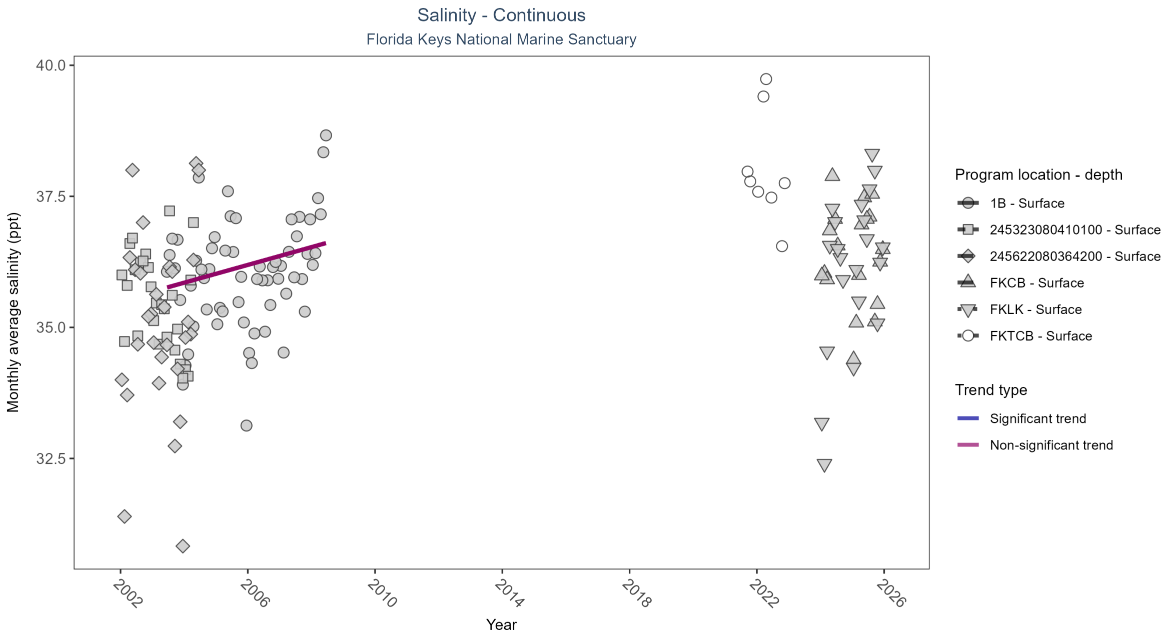

Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for five locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
1B	No detectable trend	86204	6	2003 - 2008	36.07	0.24	35.68	0.17	0.05
245323080410100	Insufficient data	746	3	2002 - 2004	35.00	-	-	-	-
245622080364200	Insufficient data	764	3	2002 - 2004	35.00	-	-	-	-
FKCB	Insufficient data	48788	2	2024 - 2025	36.50	-	-	-	-
FKLK	Insufficient data	54579	2	2024 - 2025	36.50	-	-	-	-
FKTCB	Insufficient data	13641	2	2021 - 2022	37.90	-	-	-	-

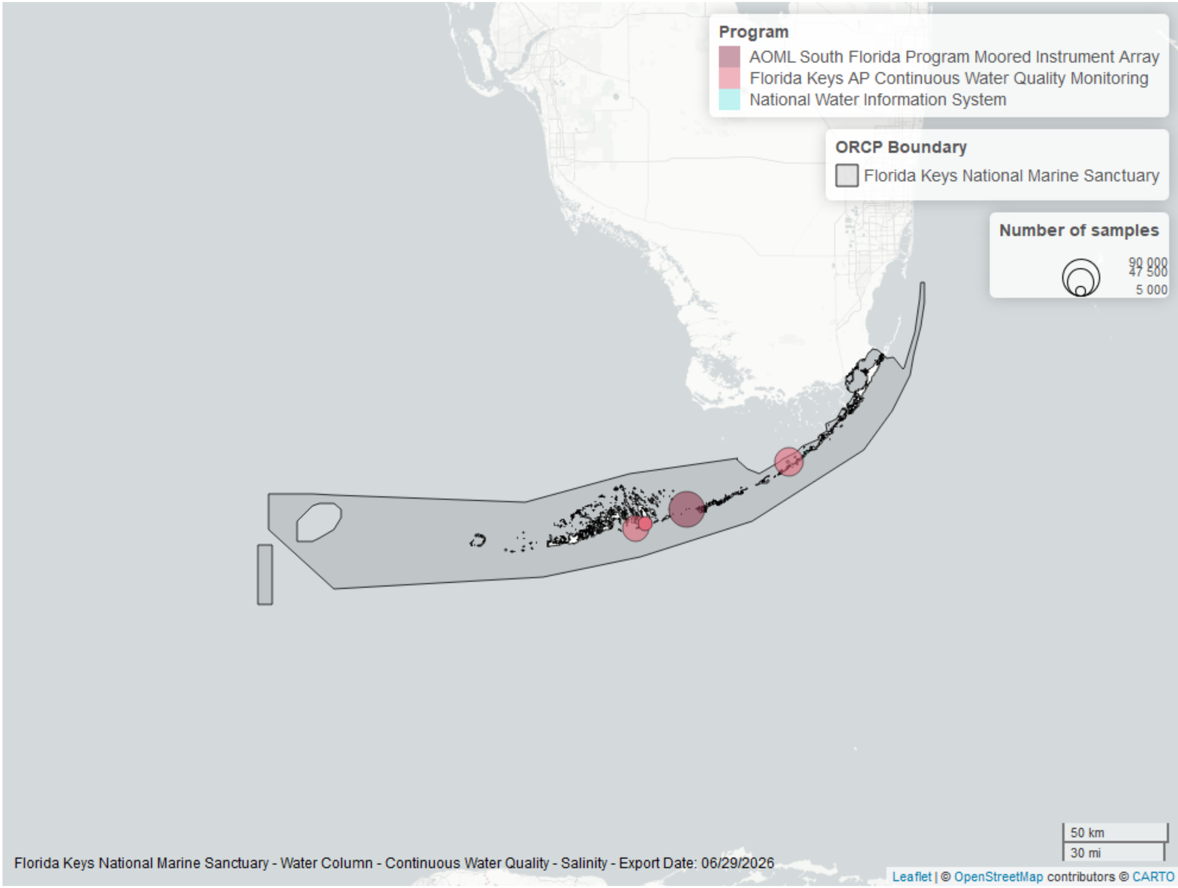

Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

Atlantic Oceanographic and Meteorological Laboratory (AOML) South Florida Program Moored Instrument Array - *2*

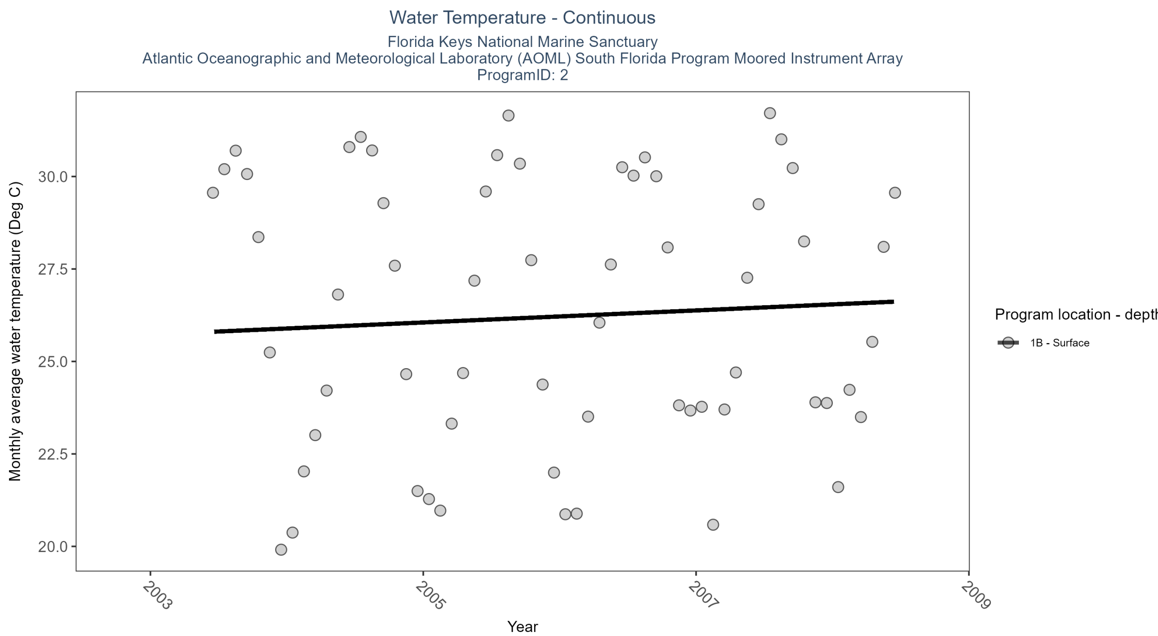

Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16Deg C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for ten locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
1B	Increasing trend	86204	6	2003 - 2008	26.38	0.26	25.73	0.16	0.04

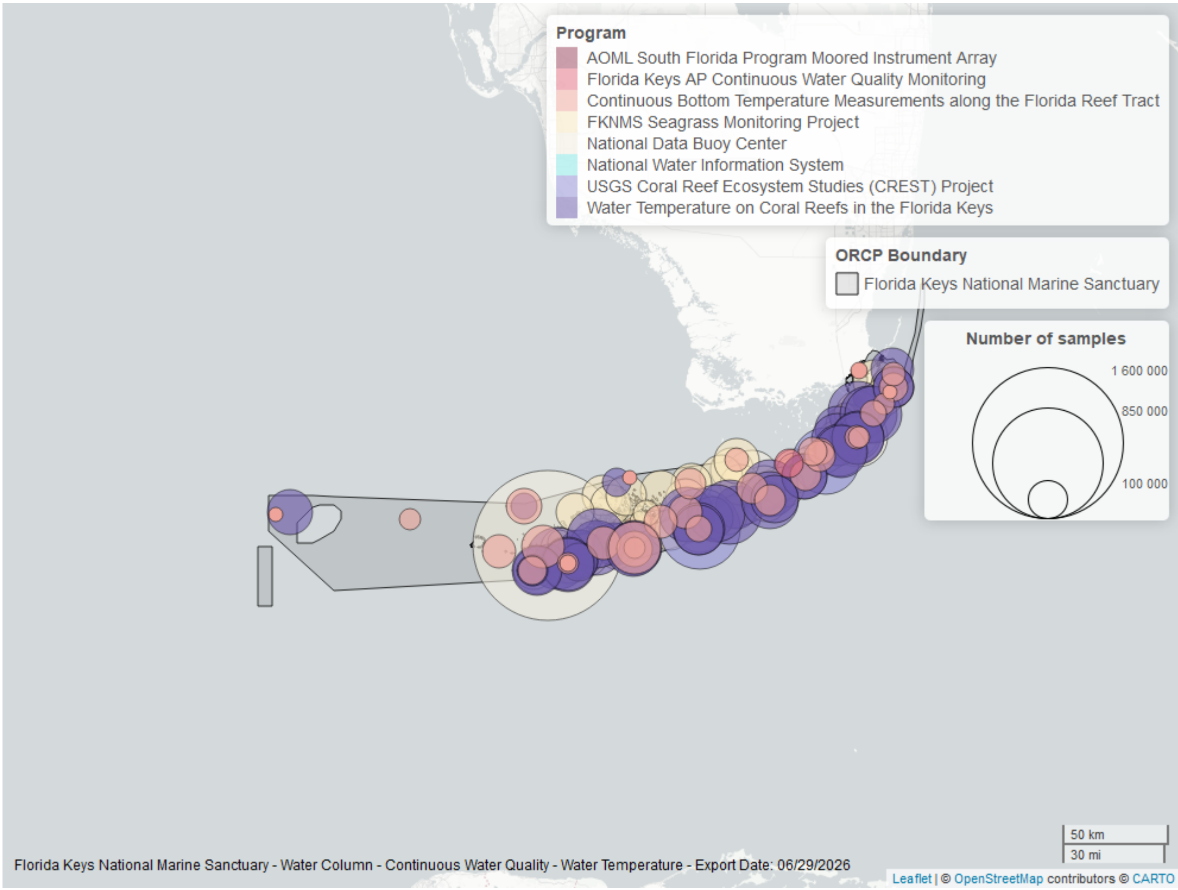

Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

National Data Buoy Center - *5*

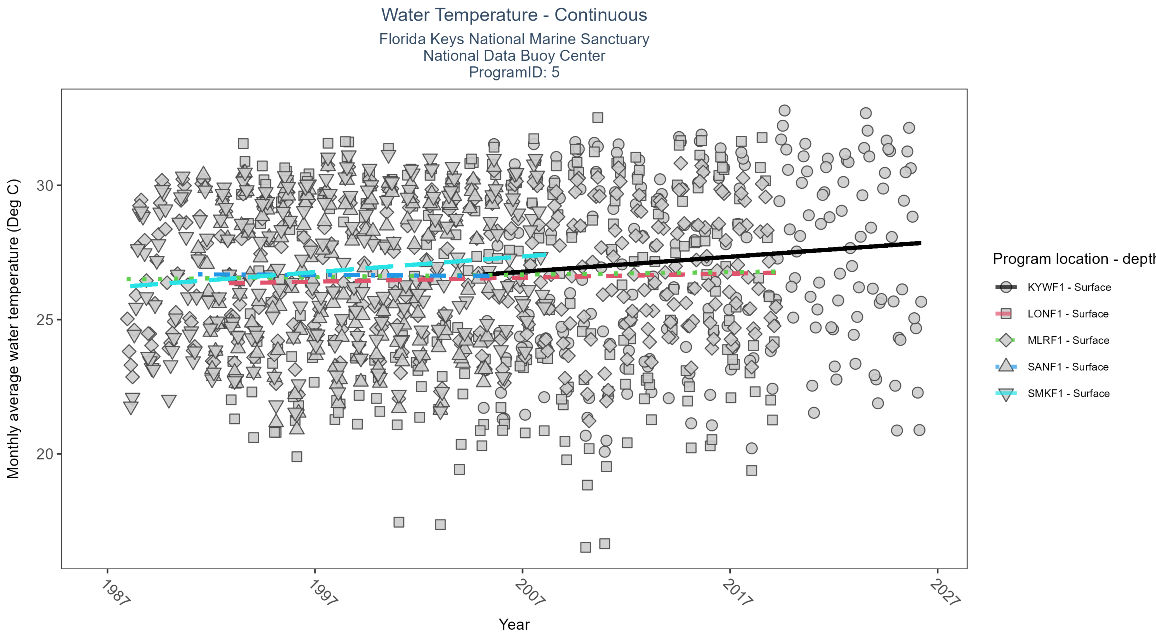

Figure 19: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16Deg C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for ten locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
KYWF1	Increasing trend	1581280	22	2005 - 2026	27.5	0.28	26.67	0.06	0.00
LONF1	No detectable trend	205971	28	1992 - 2019	26.6	0.07	26.34	0.01	0.08
MLRF1	Increasing trend	256798	33	1987 - 2019	26.5	0.10	26.49	0.01	0.00
SANF1	No detectable trend	117833	15	1991 - 2005	26.7	-0.03	26.69	0.00	0.62
SMKF1	Increasing trend	154326	21	1988 - 2008	26.8	0.34	26.24	0.06	0.00

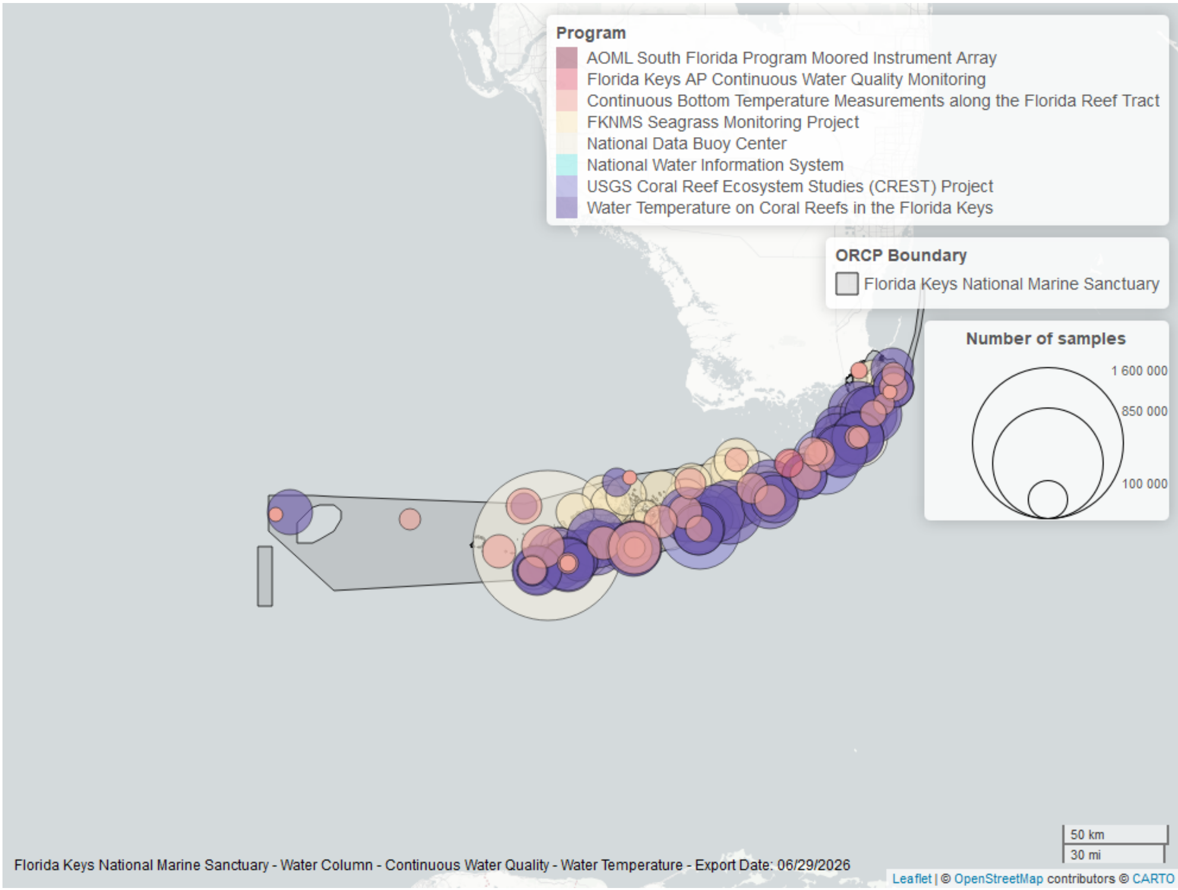

Figure 20: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

National Water Information System - *7*

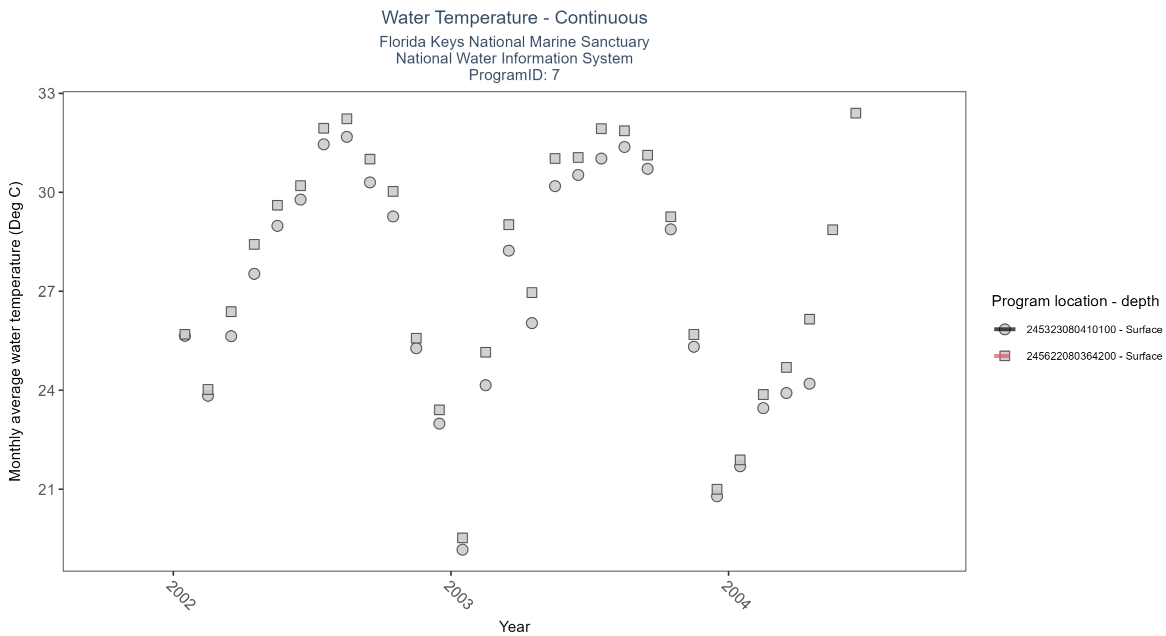

Figure 21: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16Deg C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for ten locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
245323080410100	Insufficient data	791	3	2002 - 2004	27.9	-	-	-	-
245622080364200	Insufficient data	853	3	2002 - 2004	28.3	-	-	-	-

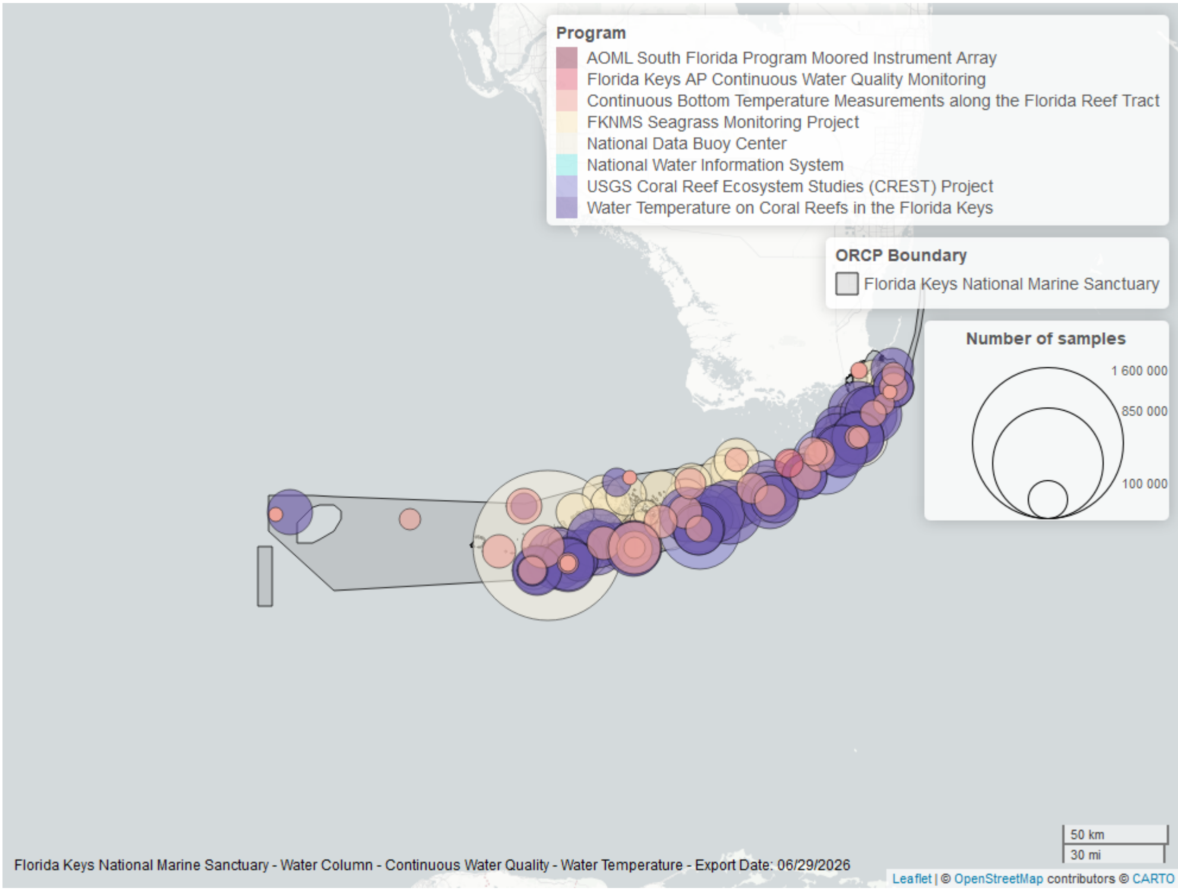

Figure 22: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Florida Keys National Marine Sanctuary Seagrass Monitoring Project - ***296***

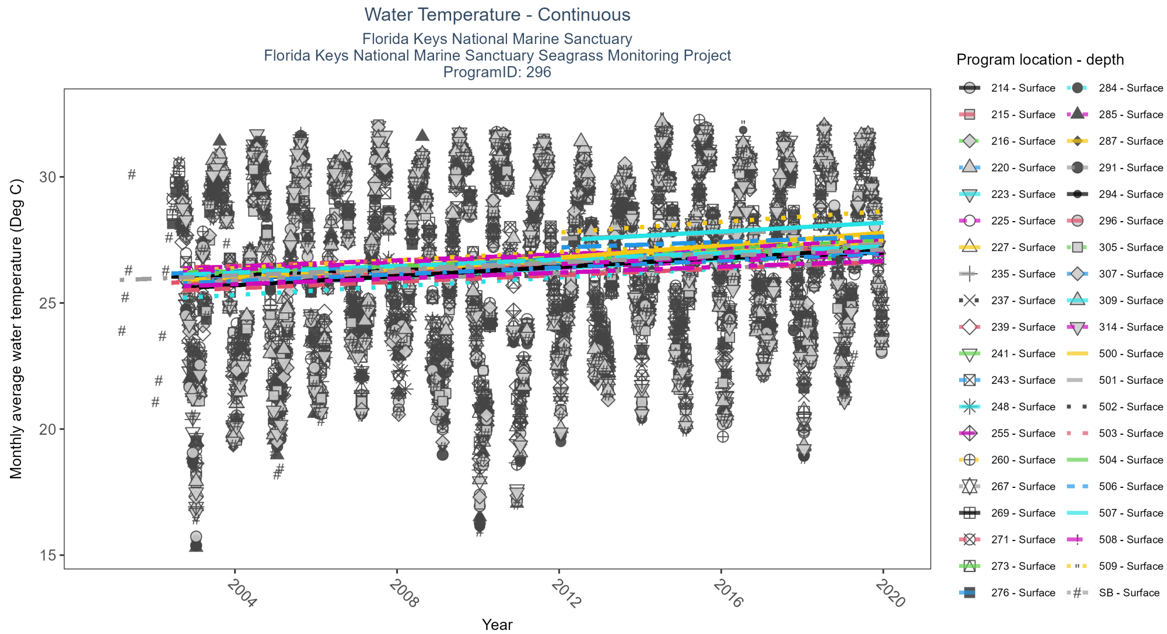

Figure 23: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16Deg C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for ten locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
214	Increasing trend	136333	18	2002 - 2019	26.52	0.27	25.84	0.07	0
215	Increasing trend	133286	16	2003 - 2018	26.74	0.26	26.42	0.05	0
216	Increasing trend	98535	17	2002 - 2018	26.26	0.31	25.86	0.06	0
220	Increasing trend	126033	17	2003 - 2019	26.52	0.25	25.94	0.06	0
223	Increasing trend	133082	18	2002 - 2019	26.89	0.3	25.84	0.08	0
225	Increasing trend	117692	17	2002 - 2019	26.82	0.32	26.32	0.06	0
227	Increasing trend	105351	17	2003 - 2019	26.67	0.29	26.06	0.08	0
235	Increasing trend	128499	18	2002 - 2019	27.14	0.28	25.77	0.08	0
237	Increasing trend	122250	18	2002 - 2019	26.38	0.31	25.74	0.09	0
239	Increasing trend	111523	17	2002 - 2018	26.92	0.24	25.96	0.07	0
241	Increasing trend	127914	18	2002 - 2019	27.26	0.27	25.91	0.09	0
243	Increasing trend	121593	18	2002 - 2019	26.62	0.3	26	0.07	0
248	Increasing trend	111702	18	2002 - 2019	26.79	0.31	25.54	0.08	0
255	Increasing trend	119939	18	2002 - 2019	26.35	0.24	25.73	0.07	0
260	Increasing trend	97832	16	2002 - 2019	27.07	0.28	26.22	0.08	0
267	Increasing trend	99735	18	2002 - 2019	26.57	0.24	25.64	0.05	0
269	Increasing trend	106458	17	2002 - 2019	26.74	0.21	26.02	0.05	0
271	Increasing trend	133627	18	2002 - 2019	26.92	0.26	25.77	0.07	0
273	Increasing trend	129817	18	2002 - 2019	27.16	0.24	26.16	0.05	0
276	Increasing trend	123833	18	2002 - 2019	26.87	0.21	26.15	0.05	0
284	Increasing trend	123977	17	2002 - 2019	26.86	0.28	25.14	0.09	0
285	Increasing trend	121423	18	2002 - 2019	26.86	0.25	26.17	0.07	0
287	Increasing trend	133008	18	2002 - 2019	26.87	0.29	25.84	0.08	0
291	Increasing trend	116240	18	2002 - 2019	26.38	0.26	25.72	0.09	0
294	Increasing trend	112348	18	2002 - 2019	26.92	0.27	25.52	0.09	0
296	Increasing trend	114497	17	2002 - 2019	27.36	0.21	25.45	0.07	0
305	Increasing trend	122296	18	2002 - 2019	26.43	0.22	26.07	0.06	0
307	Increasing trend	110802	17	2002 - 2019	26.74	0.22	25.73	0.07	0
309	Increasing trend	107410	18	2002 - 2019	27.85	0.27	26.07	0.06	0
314	Increasing trend	110686	18	2002 - 2019	27.41	0.23	25.63	0.06	0
500	Increasing trend	69048	8	2012 - 2019	27.33	0.23	26.79	0.12	0.01
501	No detectable trend	34805	5	2012 - 2018	27.48	0.11	27.55	0.05	0.65
502	Insufficient data	22765	4	2016 - 2019	26.70	-	-	-	-
503	Insufficient data	7490	1	2016 - 2016	28.74	-	-	-	-
504	Insufficient data	4339	1	2018 - 2018	29.84	-	-	-	-
506	No detectable trend	35198	7	2012 - 2019	27.41	0.04	27.2	0.05	0.74
507	No detectable trend	47517	8	2012 - 2019	27.36	0.18	27.48	0.09	0.12
508	No detectable trend	24021	6	2012 - 2019	26.67	0.33	26.54	0.07	0.29
509	No detectable trend	38607	8	2012 - 2019	27.70	0.05	27.79	0.11	0.47
SB	Increasing trend	145514	19	2001 - 2019	26.34	0.23	25.9	0.06	0

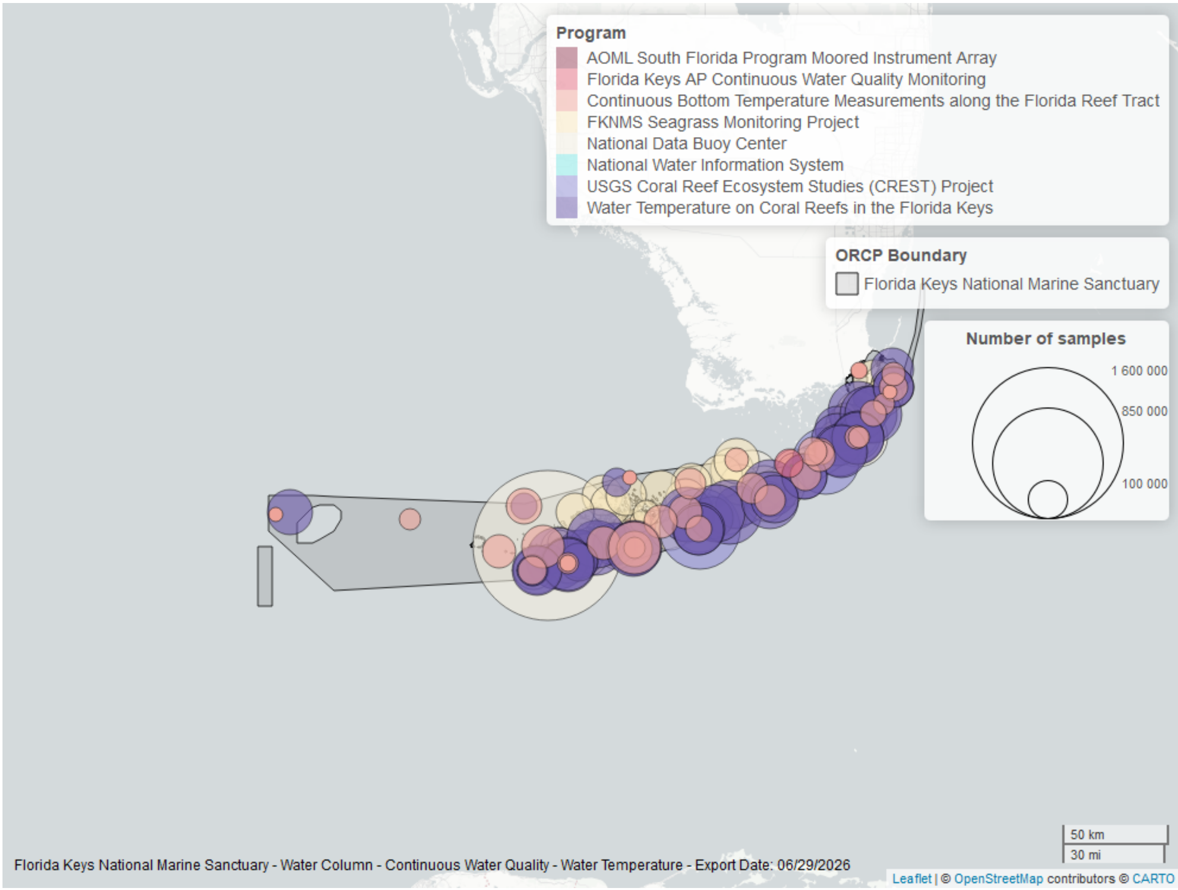

Figure 24: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

USGS Coral Reef Ecosystem Studies (CREST) Project - ***899***

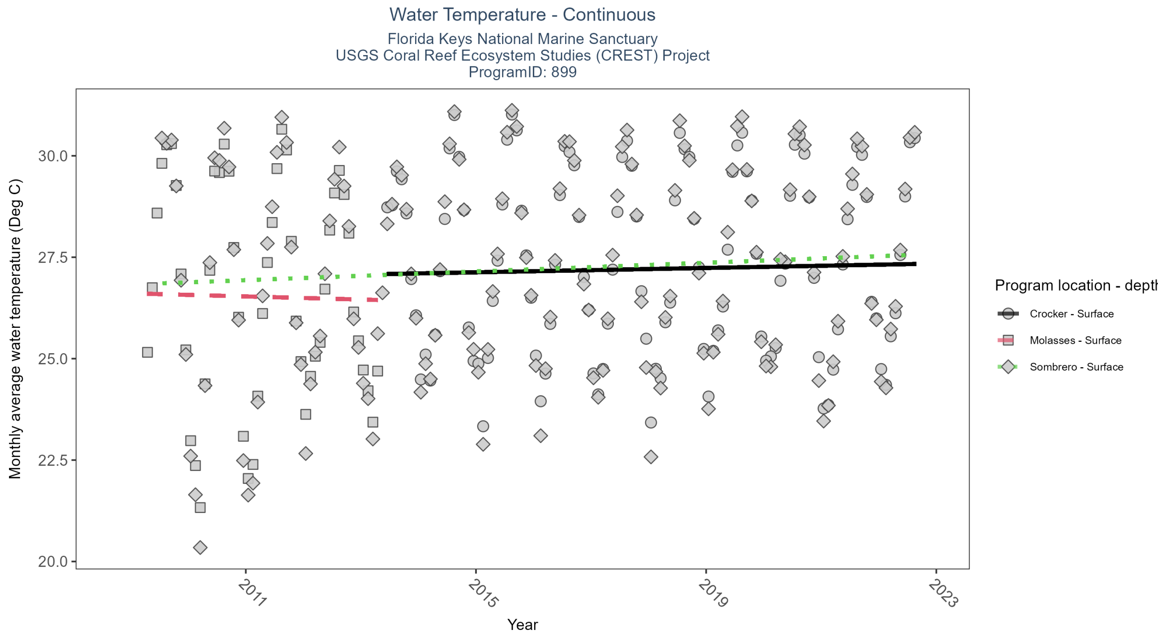

Figure 25: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16Deg C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for ten locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Crocker	Increasing trend	322670	10	2013 - 2022	27.32	0.15	27.07	0.03	0.04
Molasses	No detectable trend	140713	5	2009 - 2013	26.72	-0.03	26.61	-0.04	0.92
Sombrero	Increasing trend	459354	14	2009 - 2022	27.16	0.26	26.83	0.05	0.00

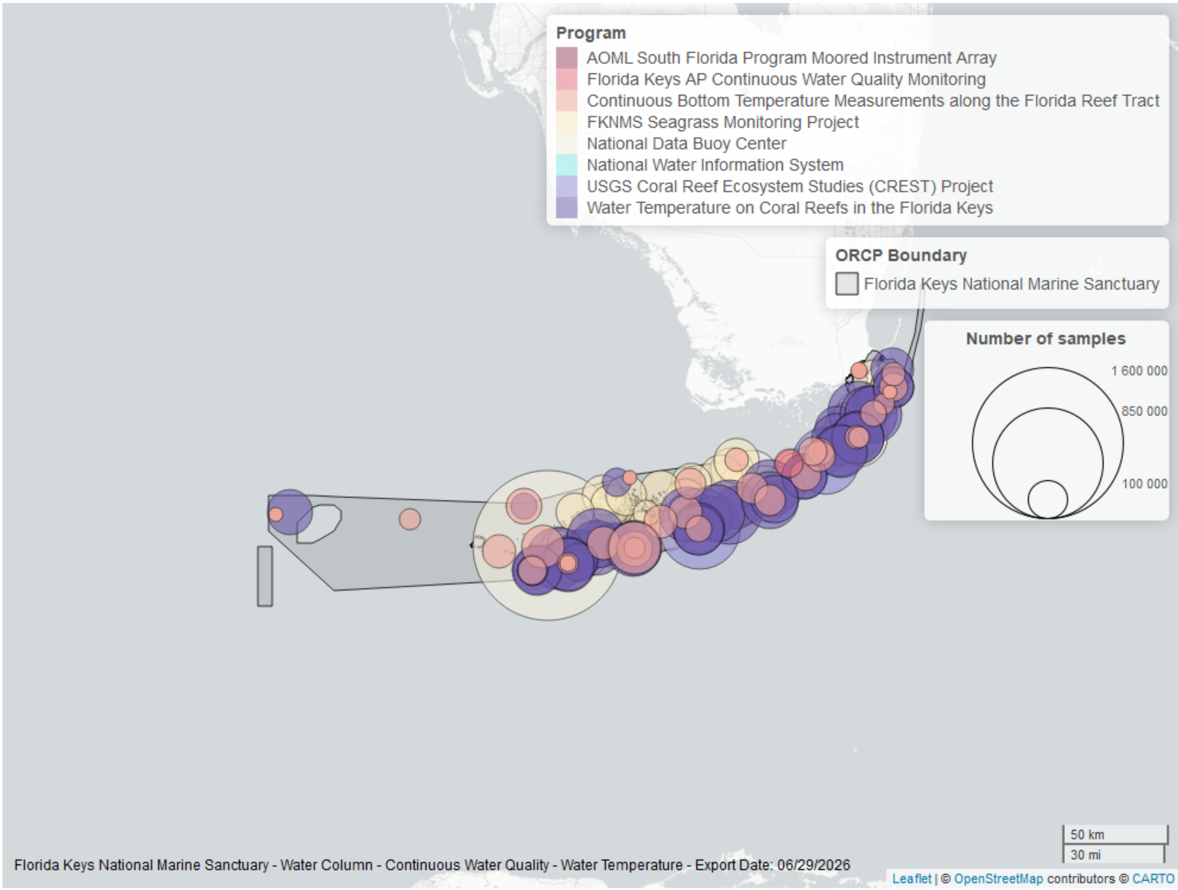

Figure 26: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature on Coral Reefs in the Florida Keys - *986*

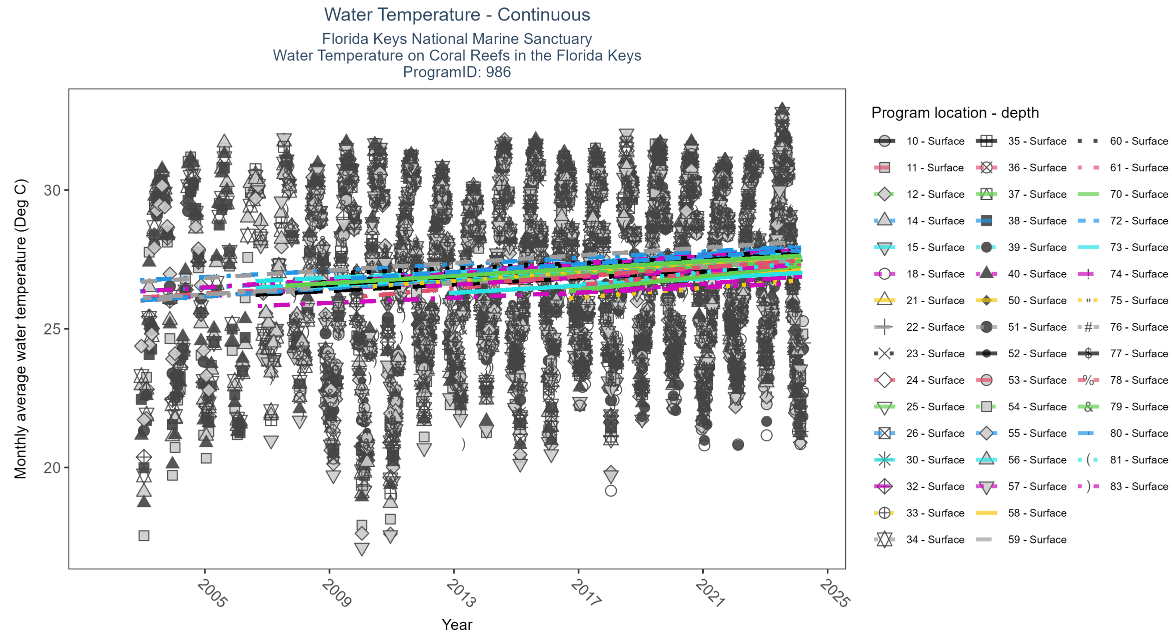

Figure 27: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16Deg C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for ten locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
10	No detectable trend	32760	5	2020 - 2024	27.48	0.23	26.68	0.20	0.18
11	Increasing trend	243135	20	2003 - 2024	26.84	0.34	26.13	0.07	0.00
12	Increasing trend	152036	15	2008 - 2024	27.18	0.26	26.41	0.07	0.00
14	Increasing trend	238807	21	2002 - 2024	26.96	0.31	26.29	0.06	0.00
15	Increasing trend	218855	19	2006 - 2024	27.04	0.23	26.45	0.06	0.00
18	Increasing trend	59983	9	2016 - 2024	27.18	0.20	26.21	0.08	0.03
21	Increasing trend	69835	9	2016 - 2024	27.23	0.20	26.46	0.08	0.03
22	Increasing trend	185473	16	2009 - 2024	26.96	0.28	26.35	0.07	0.00
23	Increasing trend	128184	13	2012 - 2024	27.43	0.27	26.73	0.08	0.00
24	Increasing trend	126389	13	2010 - 2024	27.08	0.40	26.16	0.10	0.00
25	Increasing trend	132181	14	2010 - 2024	27.28	0.21	26.73	0.07	0.00
26	Increasing trend	156891	16	2009 - 2024	27.08	0.31	26.74	0.08	0.00
30	Increasing trend	131074	13	2012 - 2024	26.67	0.26	26.23	0.06	0.00
32	Increasing trend	237317	20	2003 - 2024	26.74	0.36	26.04	0.07	0.00
33	No detectable trend	50595	8	2016 - 2024	27.06	0.17	26.04	0.09	0.18
34	Increasing trend	289051	23	2002 - 2024	26.84	0.36	25.94	0.07	0.00
35	Increasing trend	232782	19	2006 - 2024	26.98	0.28	26.19	0.06	0.00
36	Increasing trend	207795	18	2007 - 2024	27.01	0.32	26.43	0.07	0.00
37	Increasing trend	67333	9	2016 - 2024	27.03	0.30	26.27	0.12	0.00
38	Increasing trend	271179	23	2002 - 2024	26.59	0.35	25.96	0.07	0.00
39	No detectable trend	48939	7	2018 - 2024	27.06	0.09	27.06	0.08	0.45
40	Increasing trend	259048	23	2002 - 2024	26.89	0.33	26.28	0.07	0.00
50	Increasing trend	118231	12	2013 - 2024	27.11	0.33	26.82	0.07	0.00
51	Increasing trend	237296	20	2003 - 2024	26.74	0.37	26.11	0.06	0.00
52	Increasing trend	202154	17	2008 - 2024	27.01	0.40	26.61	0.08	0.00
53	Increasing trend	193391	17	2008 - 2024	27.06	0.42	26.53	0.08	0.00
54	Increasing trend	142926	13	2012 - 2024	27.09	0.31	26.64	0.06	0.00
55	Increasing trend	231874	23	2002 - 2024	26.94	0.33	26.69	0.06	0.00
56	Increasing trend	190431	19	2006 - 2024	26.81	0.28	26.69	0.05	0.00
57	Increasing trend	202888	17	2008 - 2024	27.08	0.38	26.61	0.08	0.00
58	Increasing trend	87137	11	2014 - 2024	27.35	0.25	26.94	0.07	0.00
59	Increasing trend	206603	20	2002 - 2024	26.96	0.37	26.63	0.07	0.00
60	Increasing trend	156278	16	2009 - 2024	27.01	0.23	26.99	0.05	0.00
61	Increasing trend	68996	9	2016 - 2024	27.35	0.39	26.46	0.11	0.00
70	Increasing trend	119048	12	2013 - 2024	27.01	0.33	26.73	0.06	0.00
72	Increasing trend	198393	16	2008 - 2023	26.86	0.47	26.39	0.08	0.00
73	Increasing trend	193376	17	2008 - 2024	26.81	0.40	26.47	0.08	0.00
74	Increasing trend	142860	13	2012 - 2024	26.94	0.31	26.52	0.06	0.00
75	Increasing trend	159540	15	2010 - 2024	27.23	0.39	26.49	0.08	0.00
76	Increasing trend	183695	16	2009 - 2024	26.98	0.35	26.63	0.07	0.00
77	Increasing trend	203309	17	2008 - 2024	27.01	0.36	26.51	0.07	0.00
78	Increasing trend	102736	11	2014 - 2024	27.16	0.27	26.66	0.08	0.00
79	Increasing trend	190275	18	2007 - 2024	26.94	0.33	26.51	0.07	0.00
80	Increasing trend	182341	16	2009 - 2024	27.03	0.33	26.77	0.07	0.00
81	Increasing trend	68931	9	2016 - 2024	27.35	0.39	26.48	0.11	0.00
83	Increasing trend	139311	17	2006 - 2023	25.86	0.19	25.79	0.05	0.00

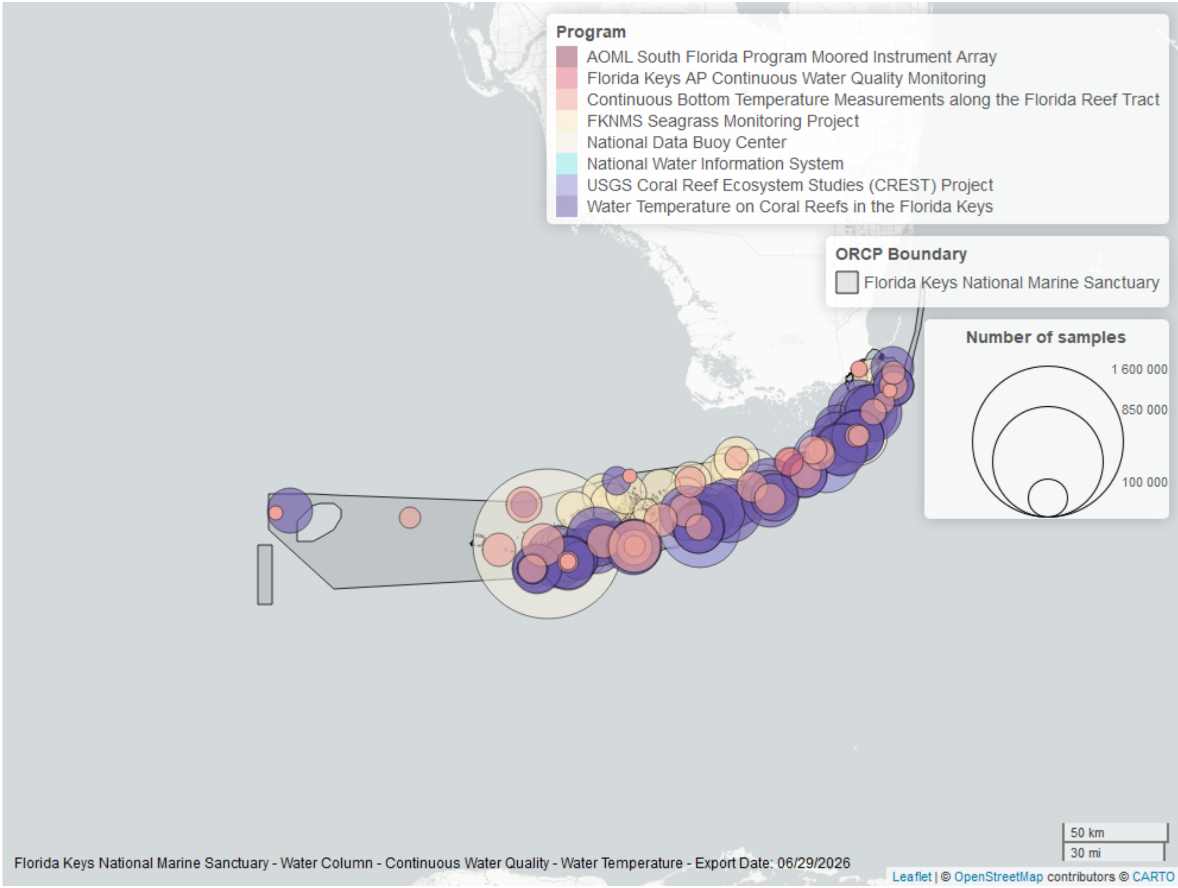

Figure 28: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Continuous Bottom Temperature Measurements along the Florida Reef Tract - *989*

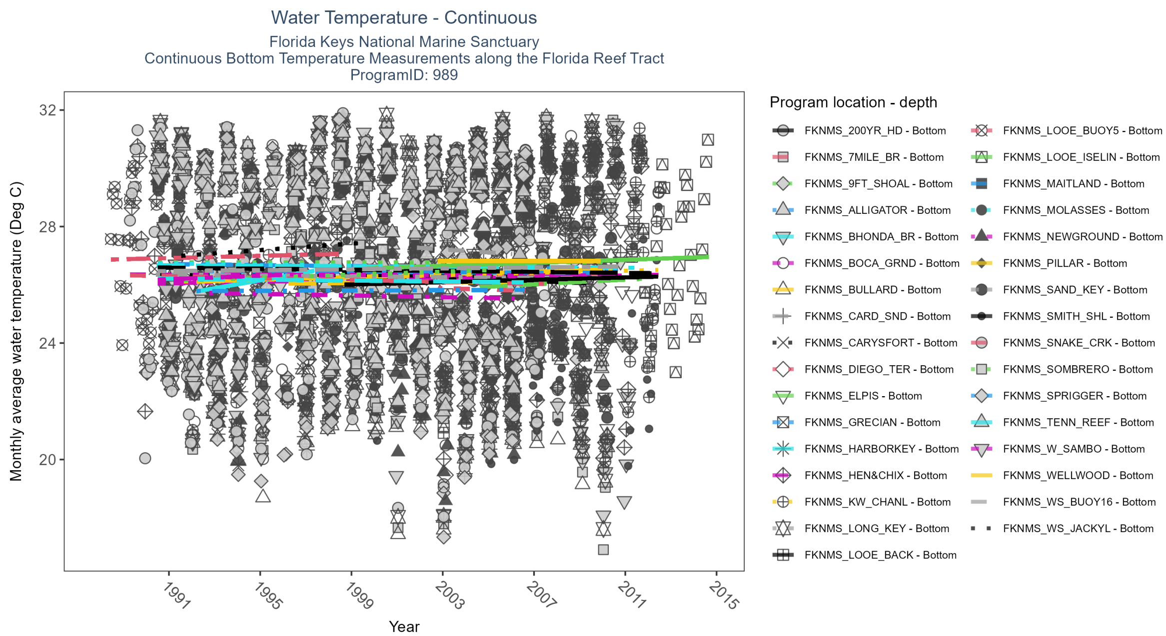

Figure 29: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16Deg C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for ten locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKNMS-200YR-HD	No detectable trend	44601	12	1998 - 2009	26.10	-0.1	26.45	-0.04	0.17
FKNMS-7MILE-BR	No detectable trend	73055	19	1991 - 2010	26.66	0.05	26.22	0.01	0.35
FKNMS-9FT-SHOAL	No detectable trend	80299	21	1990 - 2010	26.50	0	26.76	0	0.99
FKNMS-ALLIGATOR	No detectable trend	65144	19	1990 - 2010	26.55	-0.06	26.72	-0.01	0.23
FKNMS-BHONDA-BR	No detectable trend	77111	22	1990 - 2011	26.60	-0.02	26.67	0	0.66
FKNMS-BOCA-GRND	No detectable trend	73434	17	1990 - 2012	26.14	0.08	26.04	0.01	0.17
FKNMS-BULLARD	Increasing trend	66230	18	1992 - 2009	26.31	0.12	26.11	0.02	0.03
FKNMS-CARD-SND	No detectable trend	18249	6	2001 - 2006	26.52	-0.05	27.32	-0.05	0.79
FKNMS-CARYSFORT	No detectable trend	55001	16	1990 - 2006	26.40	-0.03	26.38	0	0.64
FKNMS-DIEGO-TER	No detectable trend	16693	5	2002 - 2006	25.58	-0.05	25.91	-0.03	0.84
FKNMS-ELPIS	No detectable trend	31035	8	2004 - 2011	26.35	0.06	25.9	0.04	0.53
FKNMS-GRECIAN	No detectable trend	51723	18	1990 - 2010	26.65	-0.03	26.48	0	0.66
FKNMS-HARBORKEY	No detectable trend	15407	5	1992 - 1997	26.50	0.14	25.74	0.14	0.33
FKNMS-HEN-and-CHIX	No detectable trend	72285	21	1989 - 2011	26.50	-0.01	26.35	0	0.88
FKNMS-KW-CHANL	No detectable trend	123578	18	1991 - 2012	26.27	0.1	26.11	0.02	0.08
FKNMS-LONG-KEY	No detectable trend	69656	19	1990 - 2010	26.64	-0.03	26.35	-0.01	0.58
FKNMS-LOOE-BACK	No detectable trend	84984	18	1990 - 2012	26.80	-0.06	26.6	-0.01	0.42
FKNMS-LOOE-BUOY5	No detectable trend	35252	10	1988 - 1998	26.90	0.05	26.86	0.02	0.36
FKNMS-LOOE-ISELIN	No detectable trend	194367	13	1999 - 2014	26.88	0.13	26.55	0.03	0.08
FKNMS-MAITLAND	Insufficient data	12421	4	2004 - 2007	26.07	-	-	-	-
FKNMS-MOLASSES	No detectable trend	36146	13	1990 - 2002	26.70	-0.05	26.74	-0.01	0.48
FKNMS-NEWGROUND	No detectable trend	35329	12	1992 - 2006	25.49	-0.05	25.73	-0.01	0.52
FKNMS-PILLAR	No detectable trend	40805	11	1996 - 2006	26.24	0.02	26.04	0.01	0.94
FKNMS-SAND-KEY	No detectable trend	59287	18	1990 - 2010	26.70	0.05	26.46	0.01	0.32
FKNMS-SMITH-SHL	No detectable trend	94527	10	1998 - 2012	25.45	0.13	25.99	0.02	0.19
FKNMS-SNAKE-CRK	No detectable trend	56777	19	1989 - 2007	26.16	-0.06	26.33	-0.02	0.28
FKNMS-SOMBRERO	No detectable trend	48974	13	1991 - 2005	26.50	0.13	26.14	0.03	0.05
FKNMS-SPRIGGER	No detectable trend	41834	13	1992 - 2006	26.10	0.02	25.78	0	0.86
FKNMS-TENN-REEF	No detectable trend	63260	16	1990 - 2006	26.70	-0.06	26.22	-0.01	0.27
FKNMS-WELLWOOD	No detectable trend	30427	8	2002 - 2009	26.43	0	26.82	0	1
FKNMS-WS-BUOY16	Insufficient data	8123	3	2003 - 2005	25.99	-	-	-	-
FKNMS-WS-JACKYL	No detectable trend	29557	9	1991 - 1999	26.40	0.17	26.96	0.06	0.09
FKNMS-W-SAMBO	No detectable trend	18786	6	1990 - 1995	26.90	0.09	26.16	0.03	0.56

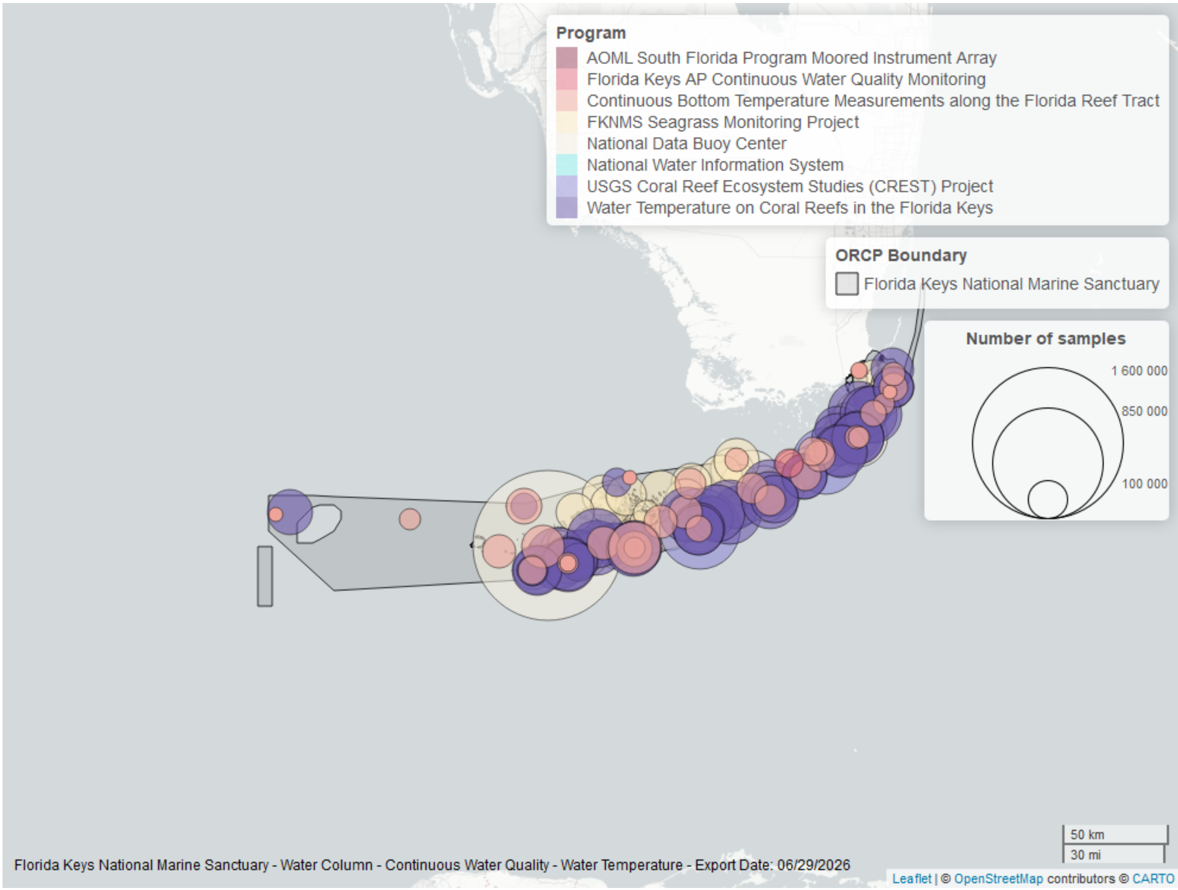

Figure 30: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

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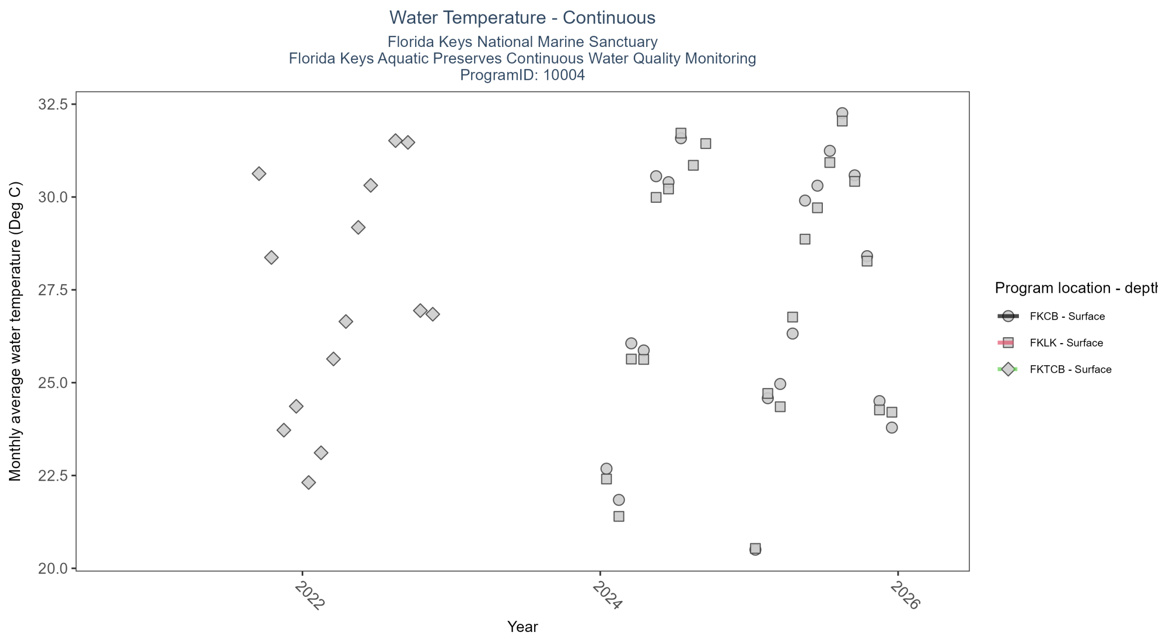

Figure 31: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 15: At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16Deg C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for ten locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	48829	2	2024 - 2025	27.4	-	-	-	-
FKLK	Insufficient data	54509	2	2024 - 2025	28.2	-	-	-	-
FKTCB	Insufficient data	30145	2	2021 - 2022	26.7	-	-	-	-

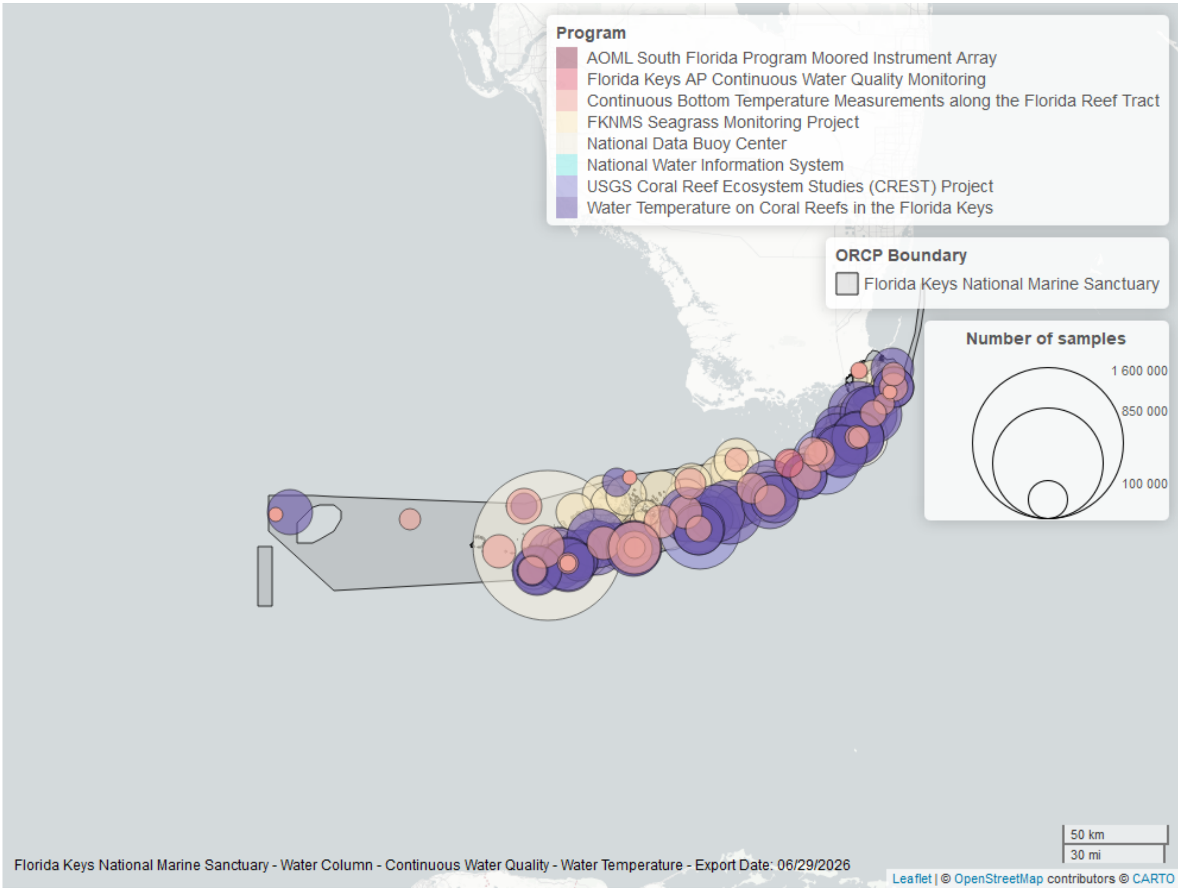

Figure 32: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

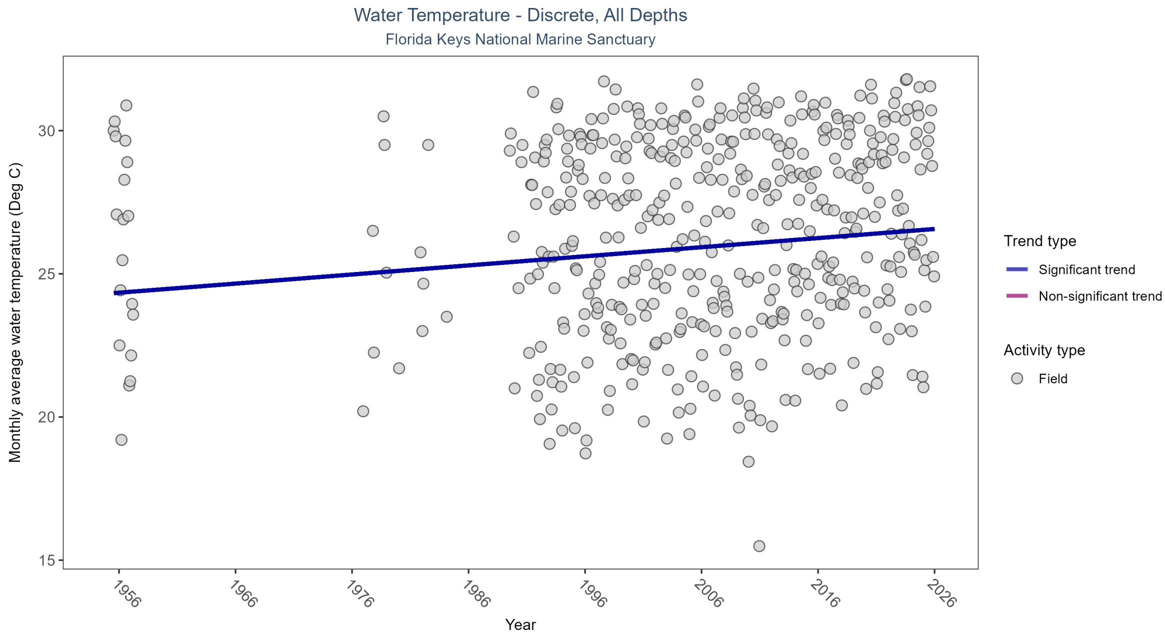

Figure 33: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 16: Monthly average water temperature increased by 0.03Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	63153	47	1955 - 2025	27.2272	0.26928	24.30906	0.03178	0

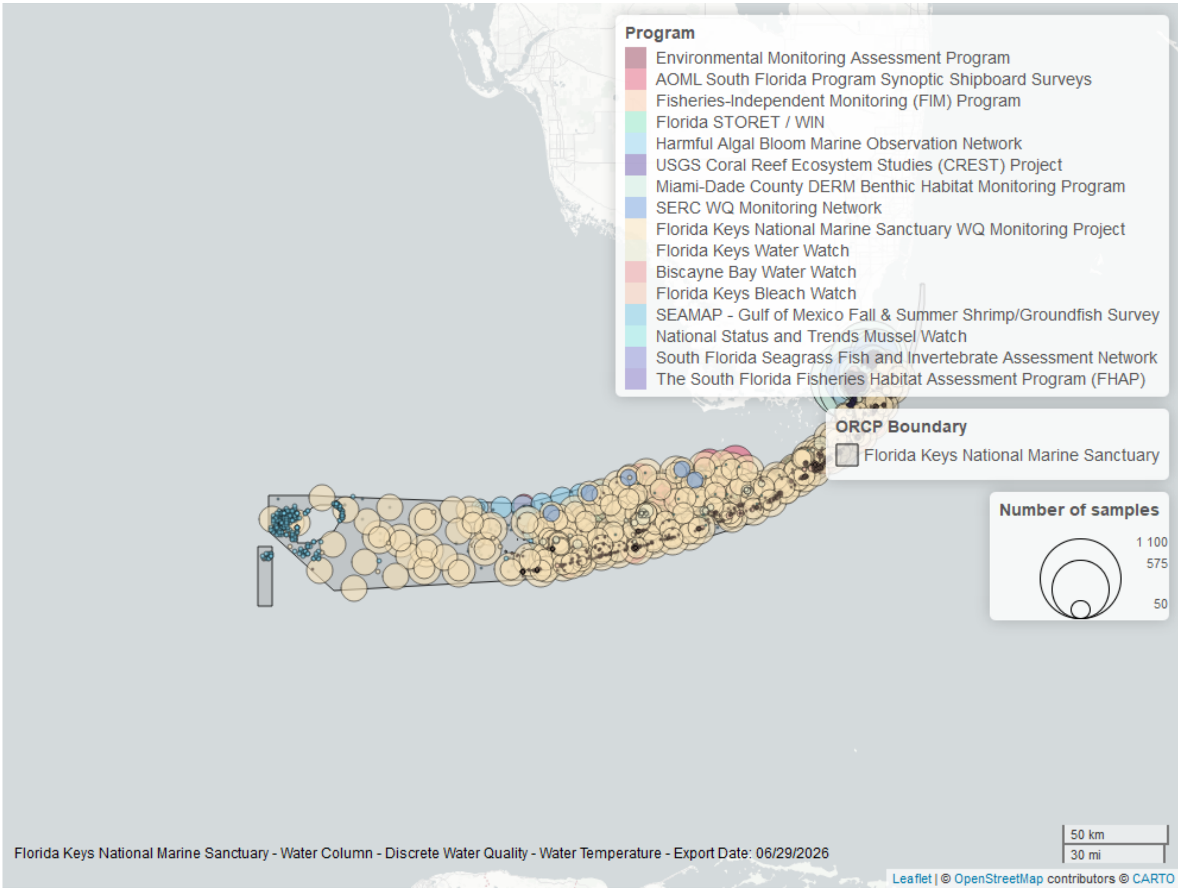

Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

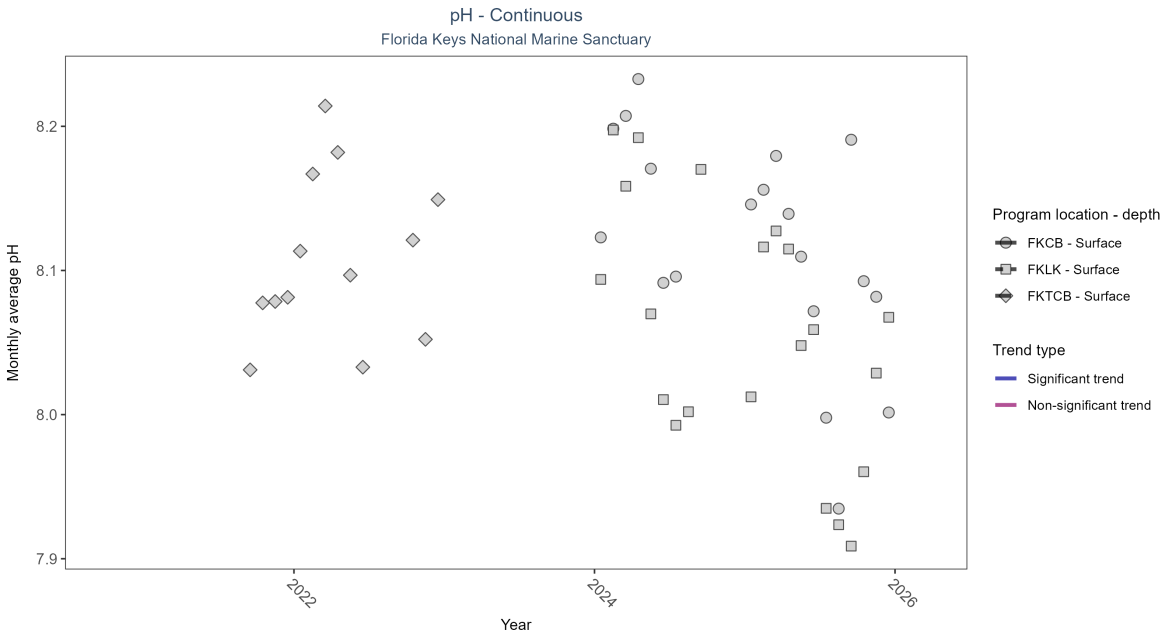

Figure 35: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 17: There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	48063	2	2024 - 2025	8.1	-	-	-	-
FKLK	Insufficient data	54596	2	2024 - 2025	8.1	-	-	-	-
FKTCB	Insufficient data	22914	2	2021 - 2022	8.1	-	-	-	-

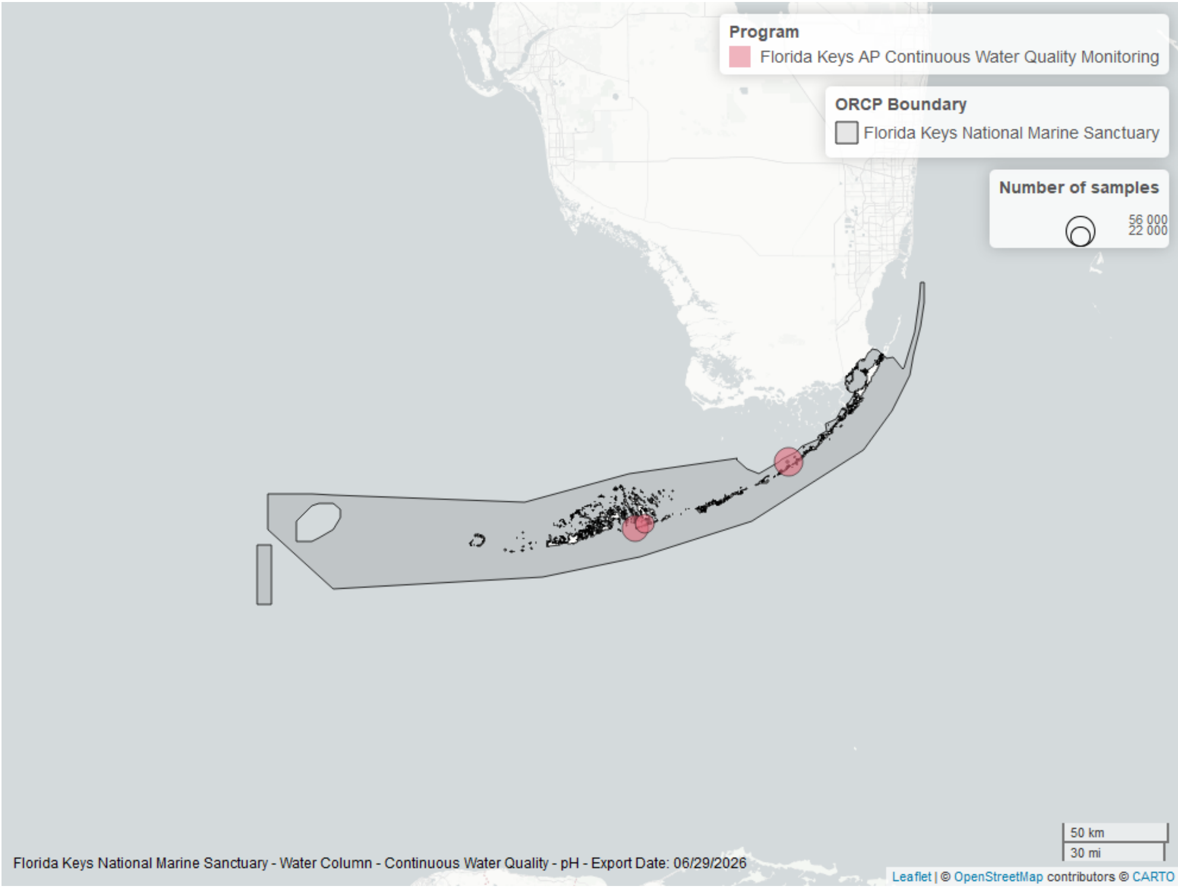

Figure 36: Map showing location of ph continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

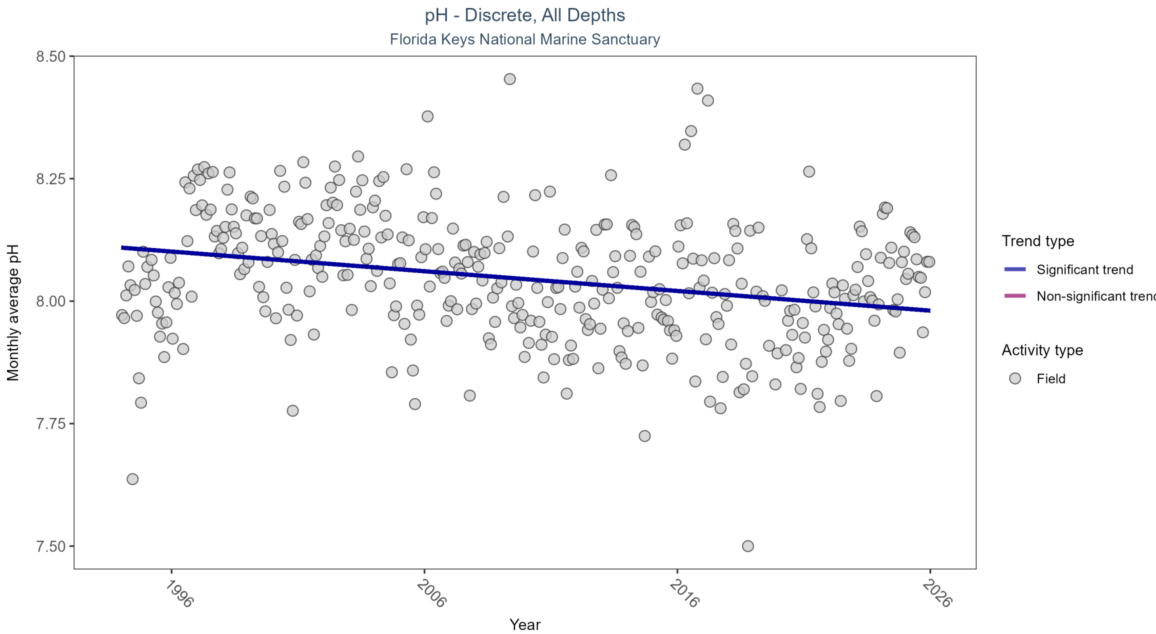

Figure 37: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 18: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	16755	32	1994 - 2025	8.081	-0.20832	8.1092	-0.00402	0

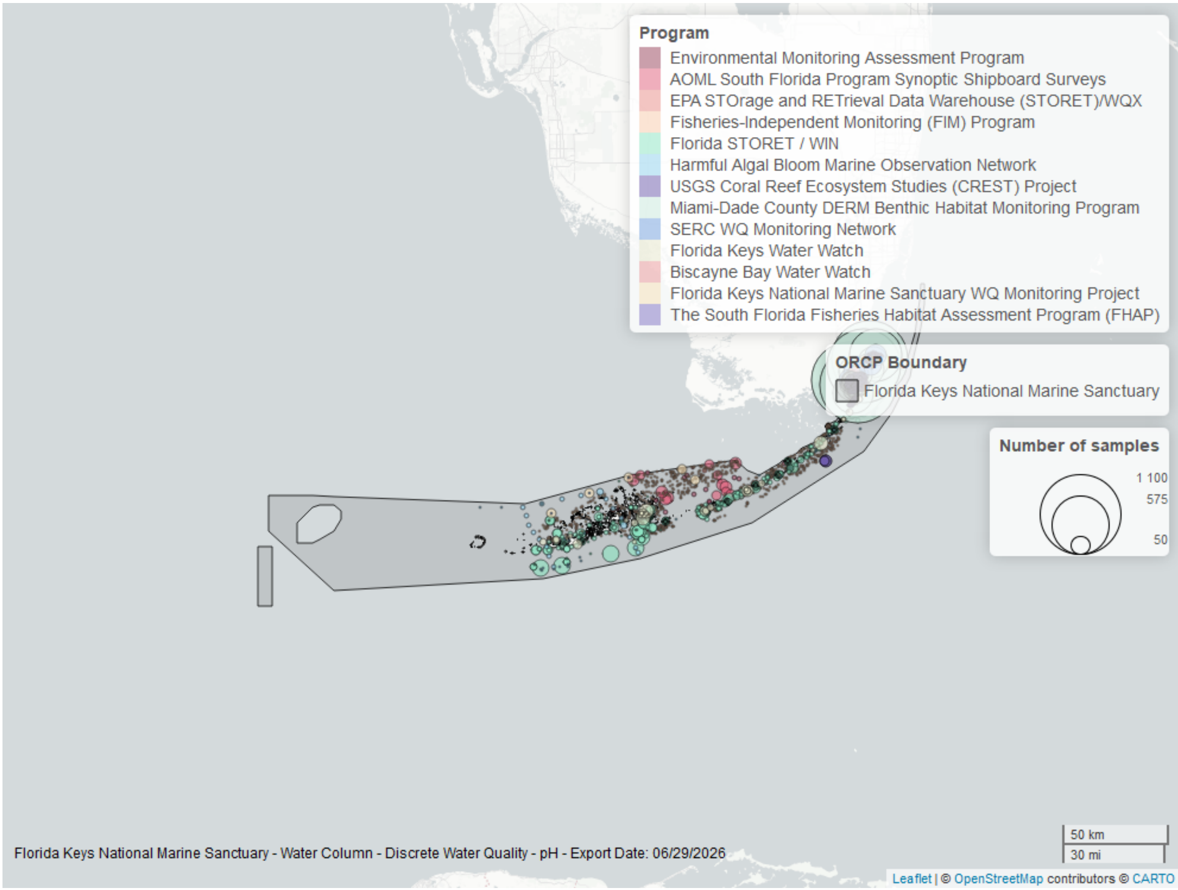

Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

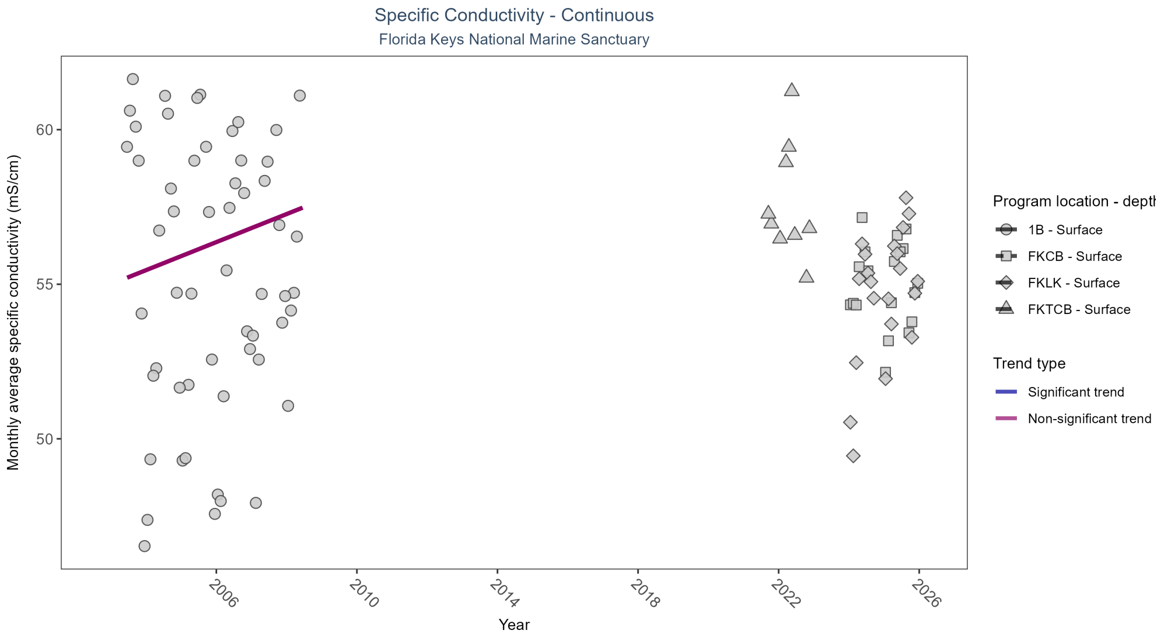

Figure 39: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 19: No detectable change in monthly average specific conductivity was observed at one location. There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
1B	No detectable trend	86204	6	2003 - 2008	56.33	0.21	55	0.45	0.1
FKCB	Insufficient data	48788	2	2024 - 2025	55.09	-	-	-	-
FKLK	Insufficient data	54579	2	2024 - 2025	55.16	-	-	-	-
FKTCB	Insufficient data	13641	2	2021 - 2022	57.01	-	-	-	-

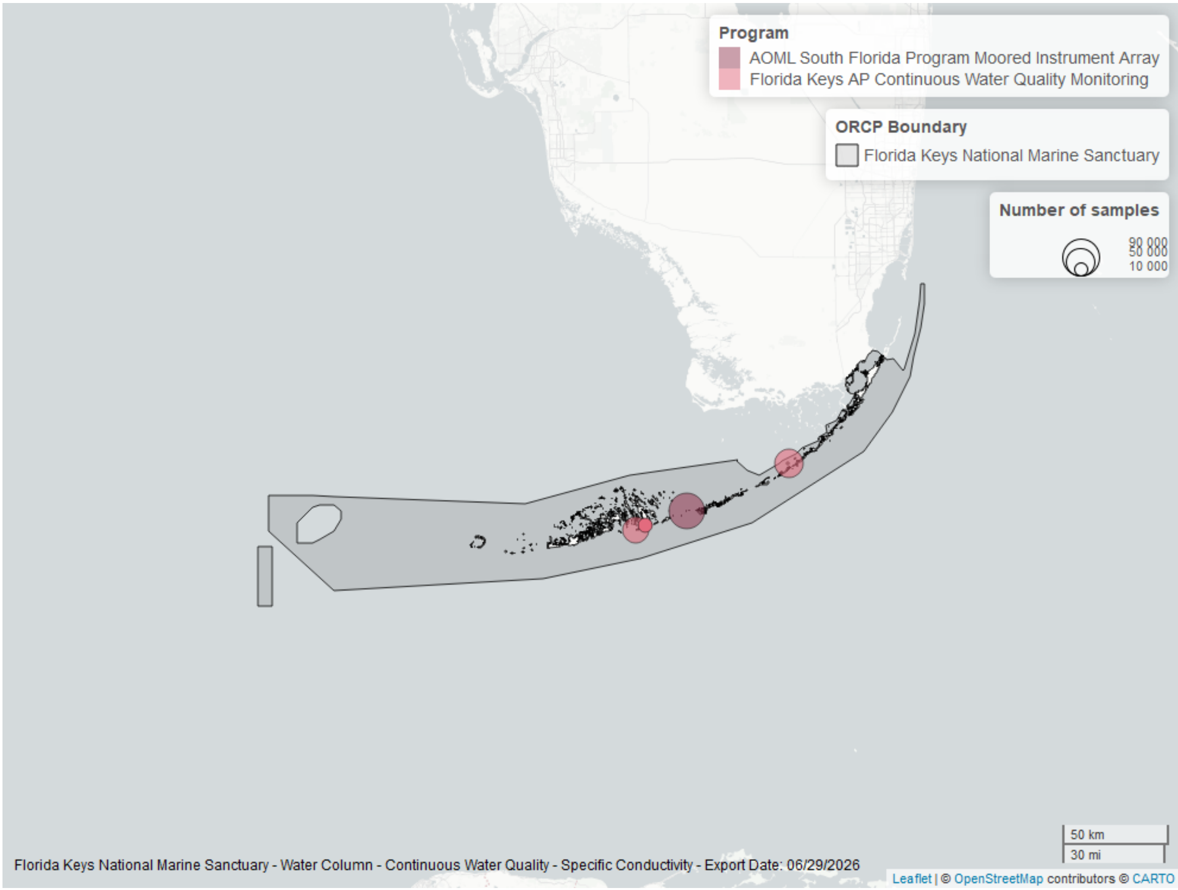

Figure 40: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

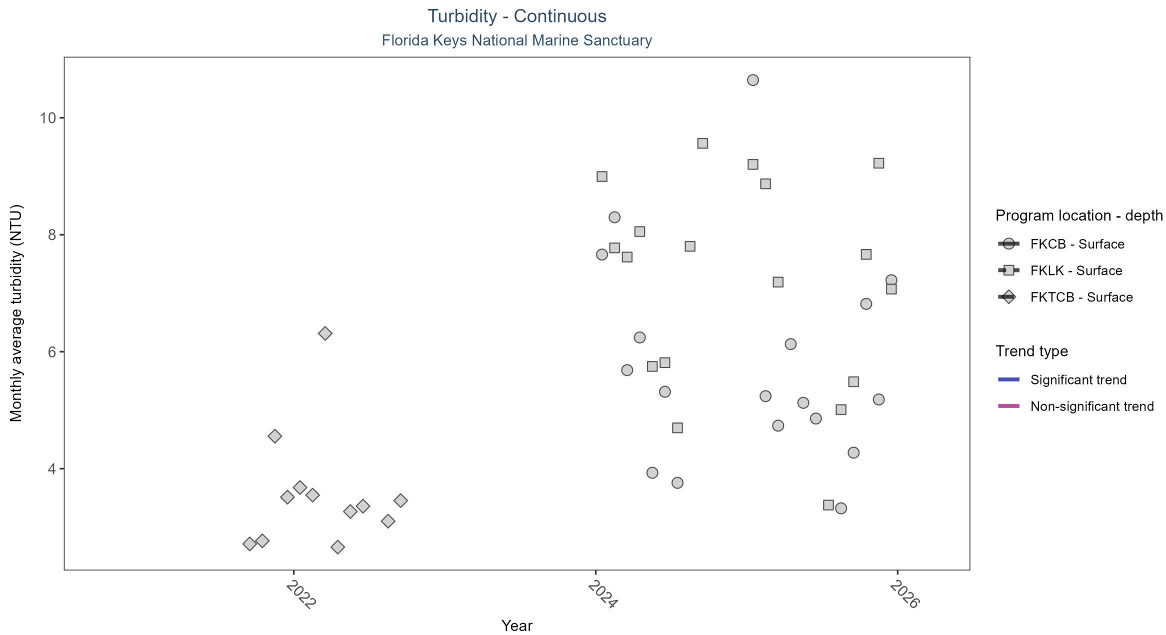

Figure 41: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 20: There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	37724	2	2024 - 2025	4	-	-	-	-
FKLK	Insufficient data	41048	2	2024 - 2025	5	-	-	-	-
FKTCB	Insufficient data	19289	2	2021 - 2022	3	-	-	-	-

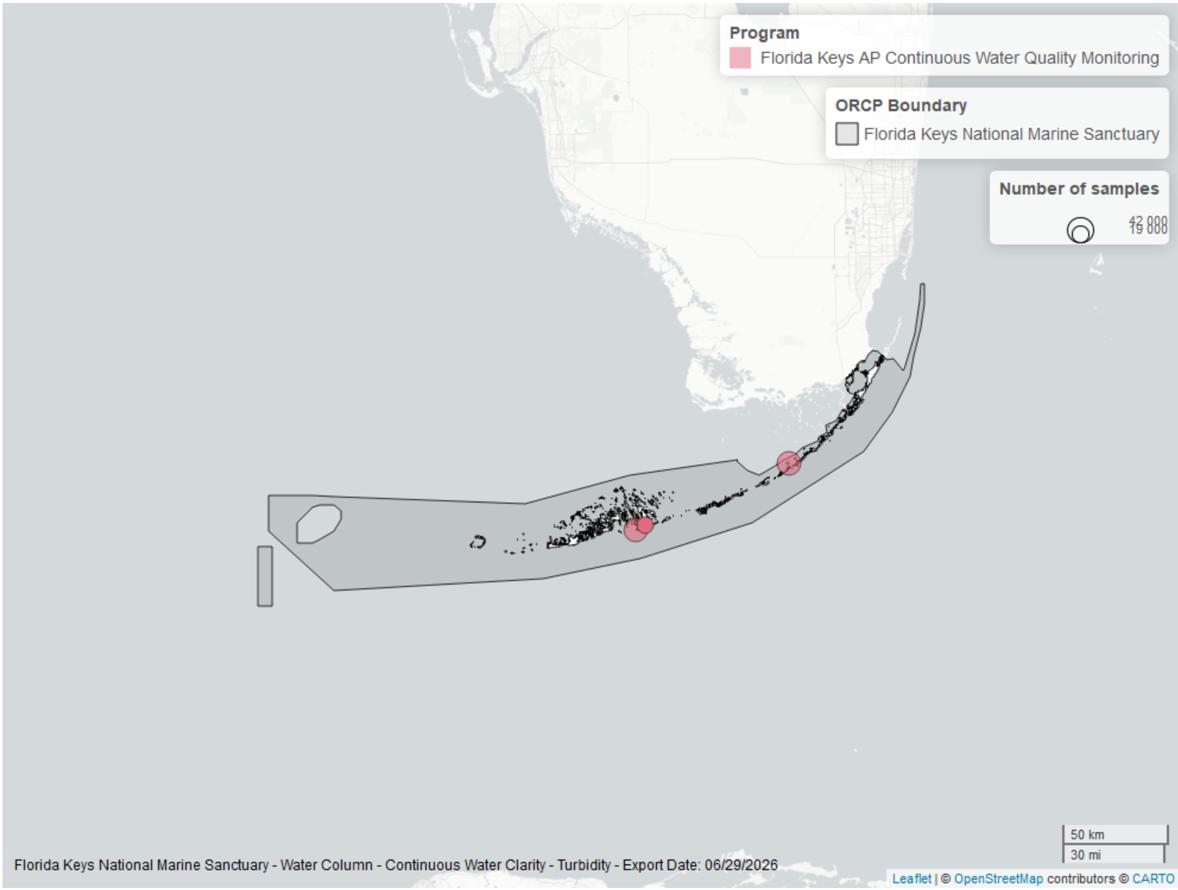

Figure 42: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

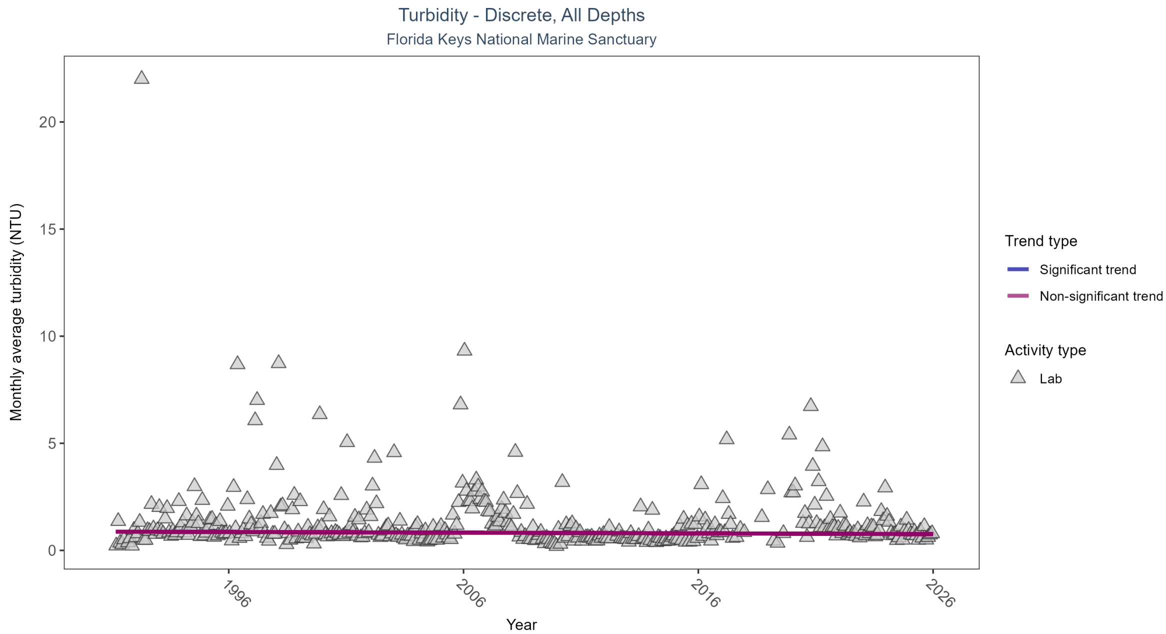

Figure 43: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 21: Turbidity showed no detectable trend between 1991 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	3974	35	1991 - 2025	0.7	-0.04939	0.87625	-0.00328	0.1706

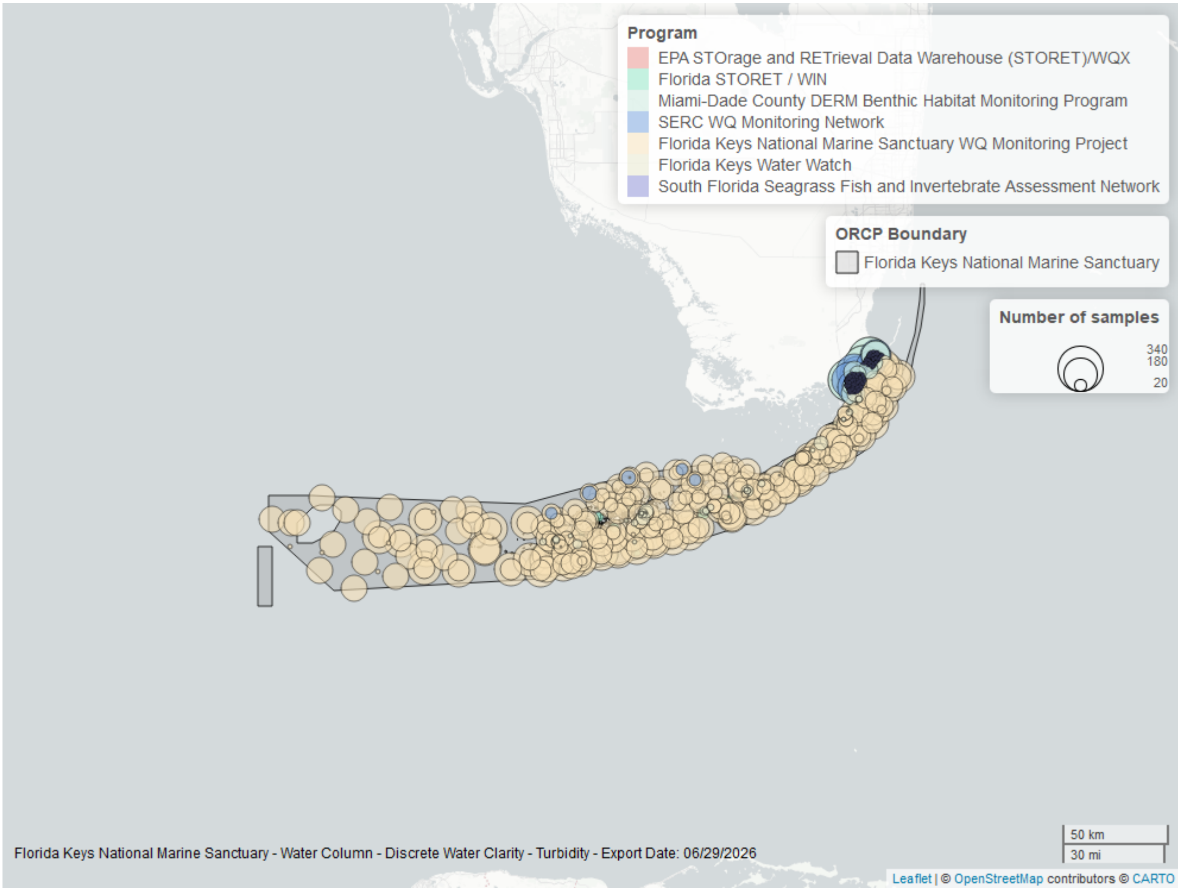

Figure 44: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

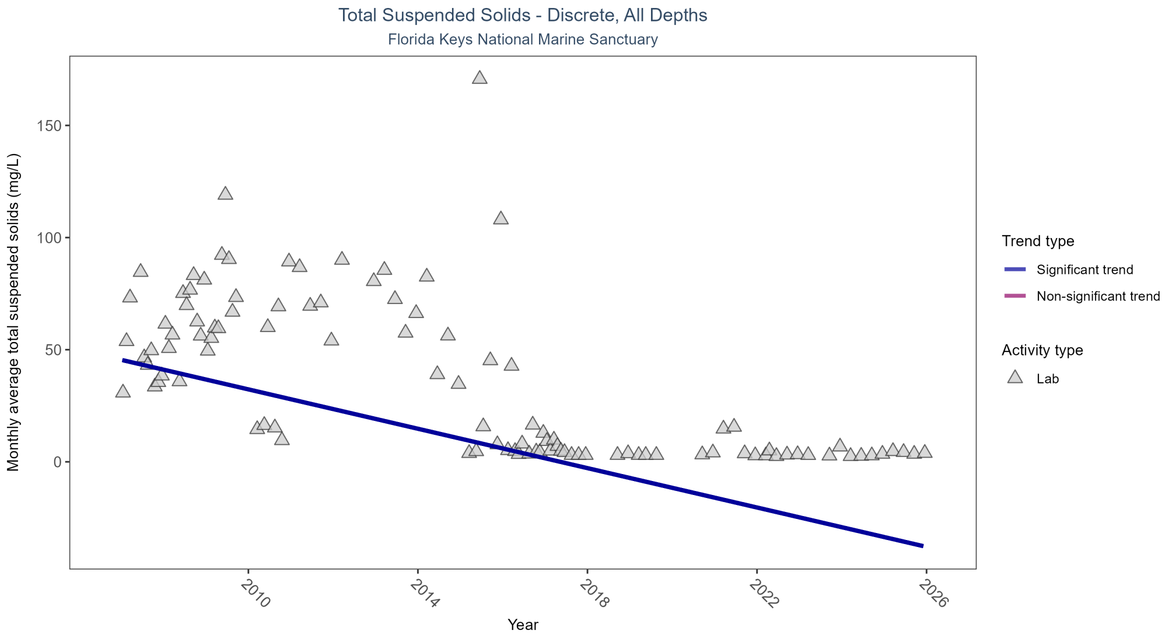

Figure 45: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Monthly average total suspended solids decreased by 4.39 mg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	560	19	2007 - 2025	13	-0.58264	45.47597	-4.38958	0

Figure 46: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

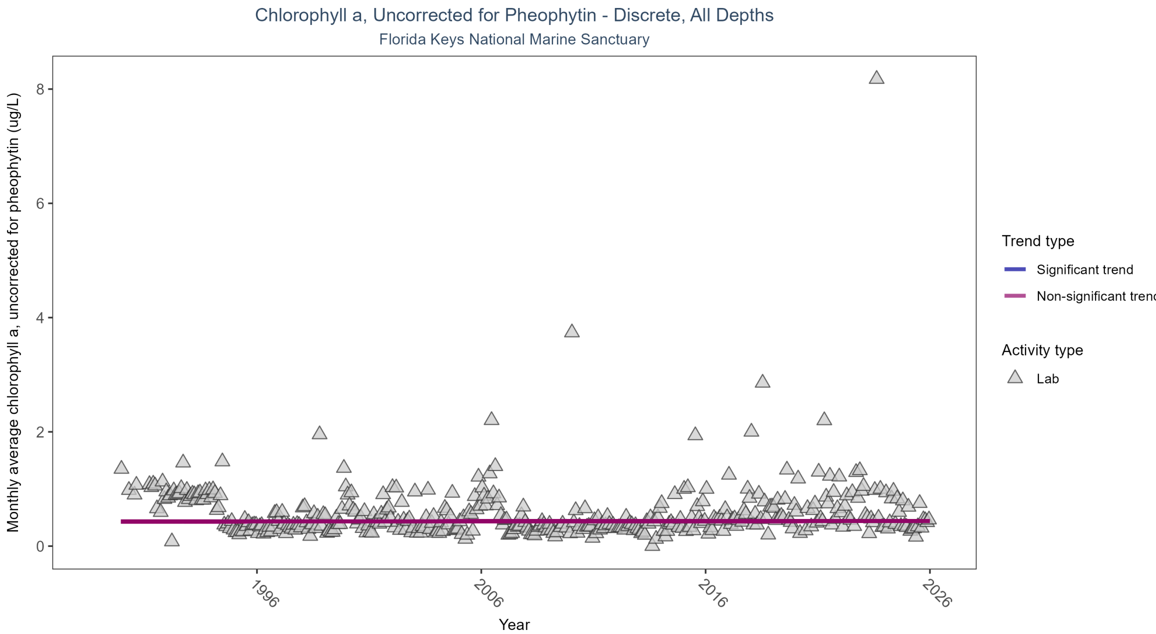

Figure 47: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 23: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 1989 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	22200	37	1989 - 2025	0.29358	0.00726	0.43012	0.00026	0.858

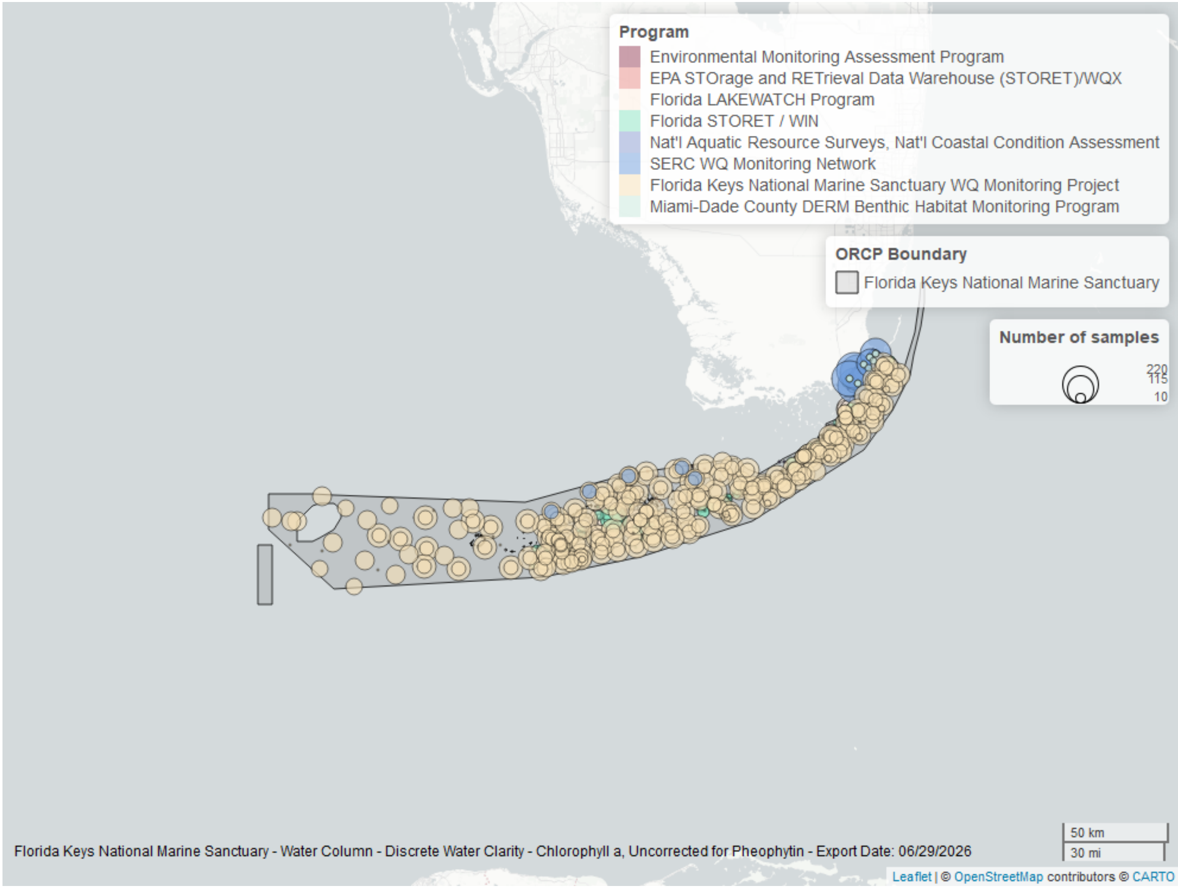

Figure 48: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

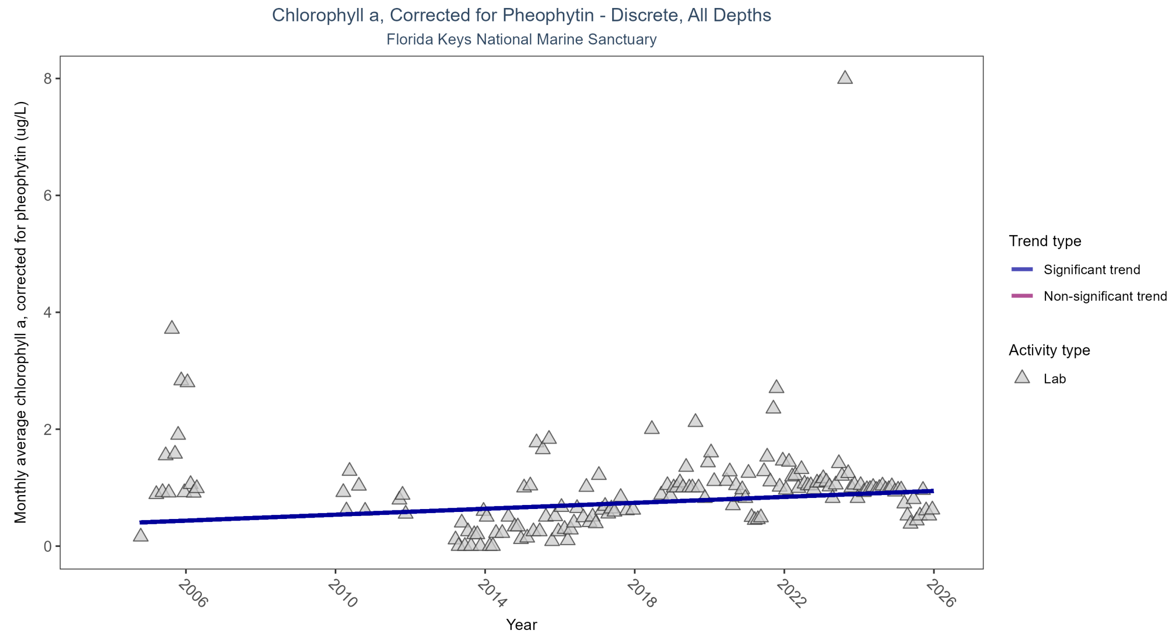

Figure 49: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 24: Monthly average chlorophyll a, corrected for pheophytin, increased by 0.03 ugg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2230	18	2004 - 2025	0.68	0.18656	0.38484	0.02539	0.0012

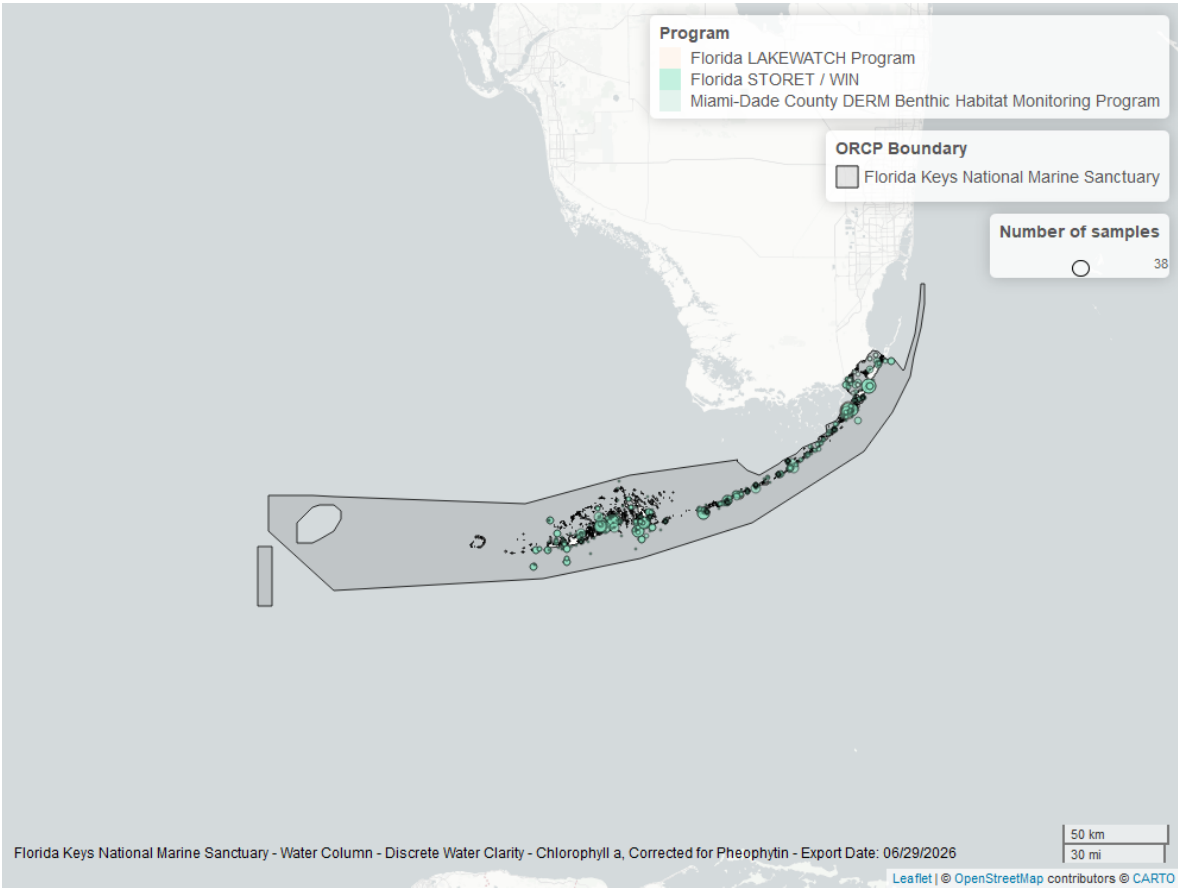

Figure 50: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

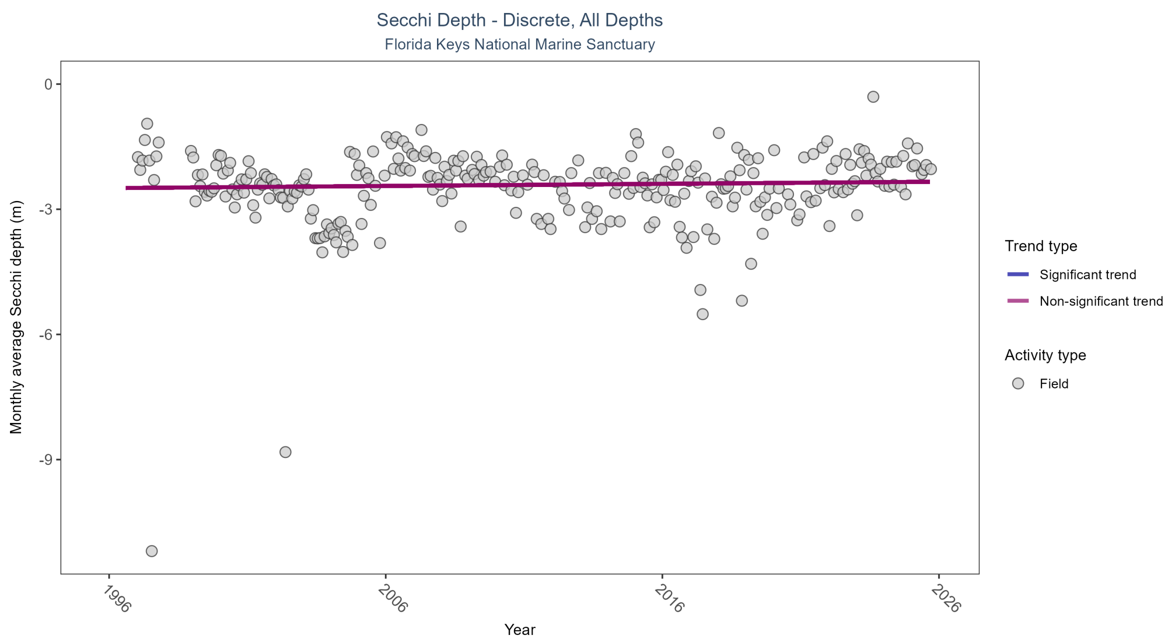

Figure 51: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 25: Secchi depth showed no detectable trend between 1996 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	5182	30	1996 - 2025	-2.13363	0.0459	-2.49352	0.00512	0.2638

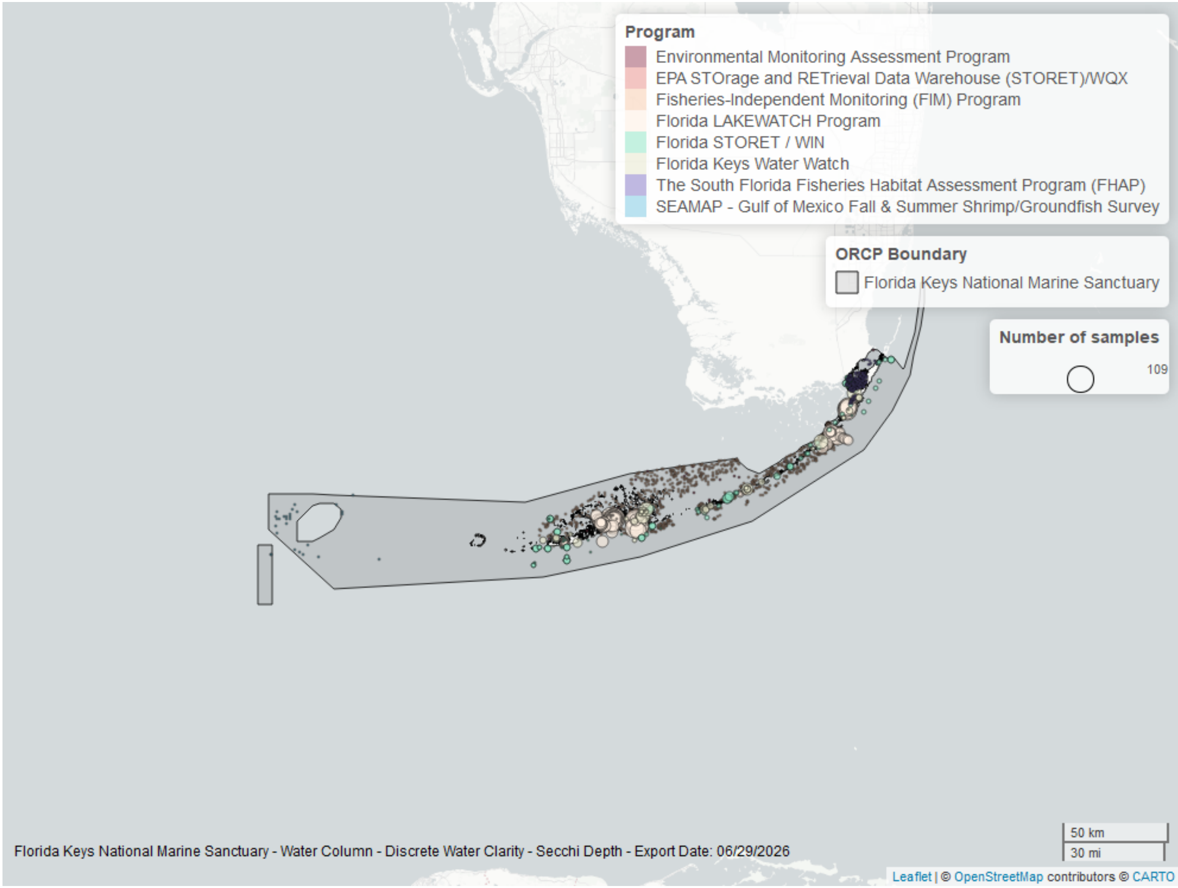

Figure 52: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

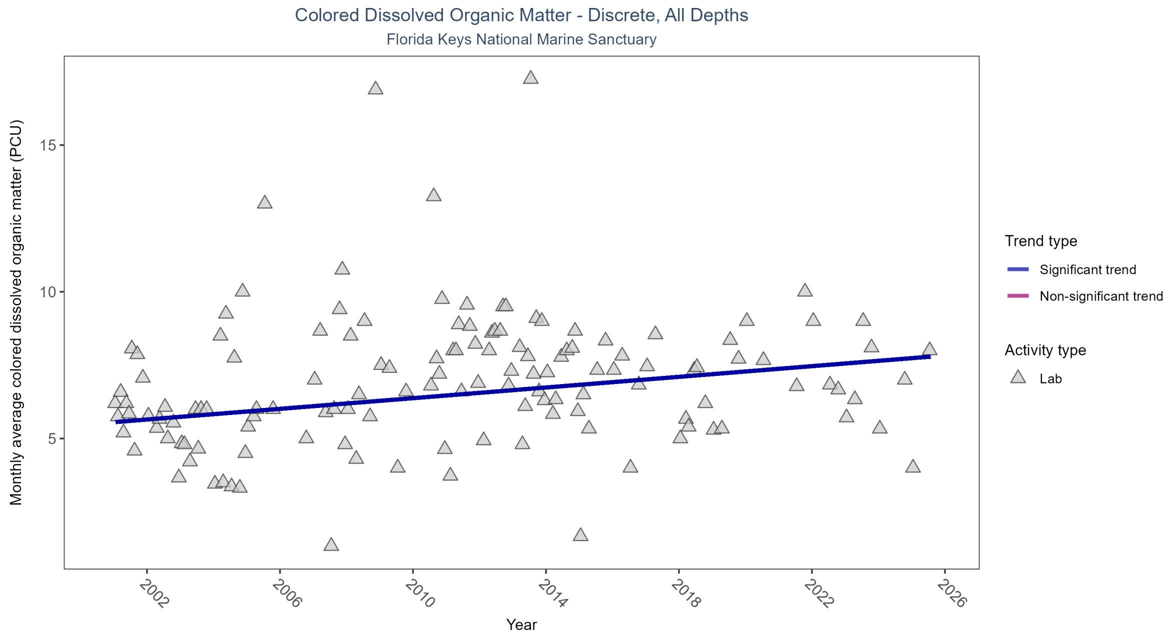

Figure 53: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 26: Monthly average colored dissolved organic matter increased by 0.09 PCU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1041	25	2001 - 2025	6	0.22711	5.5569	0.09091	0.0009

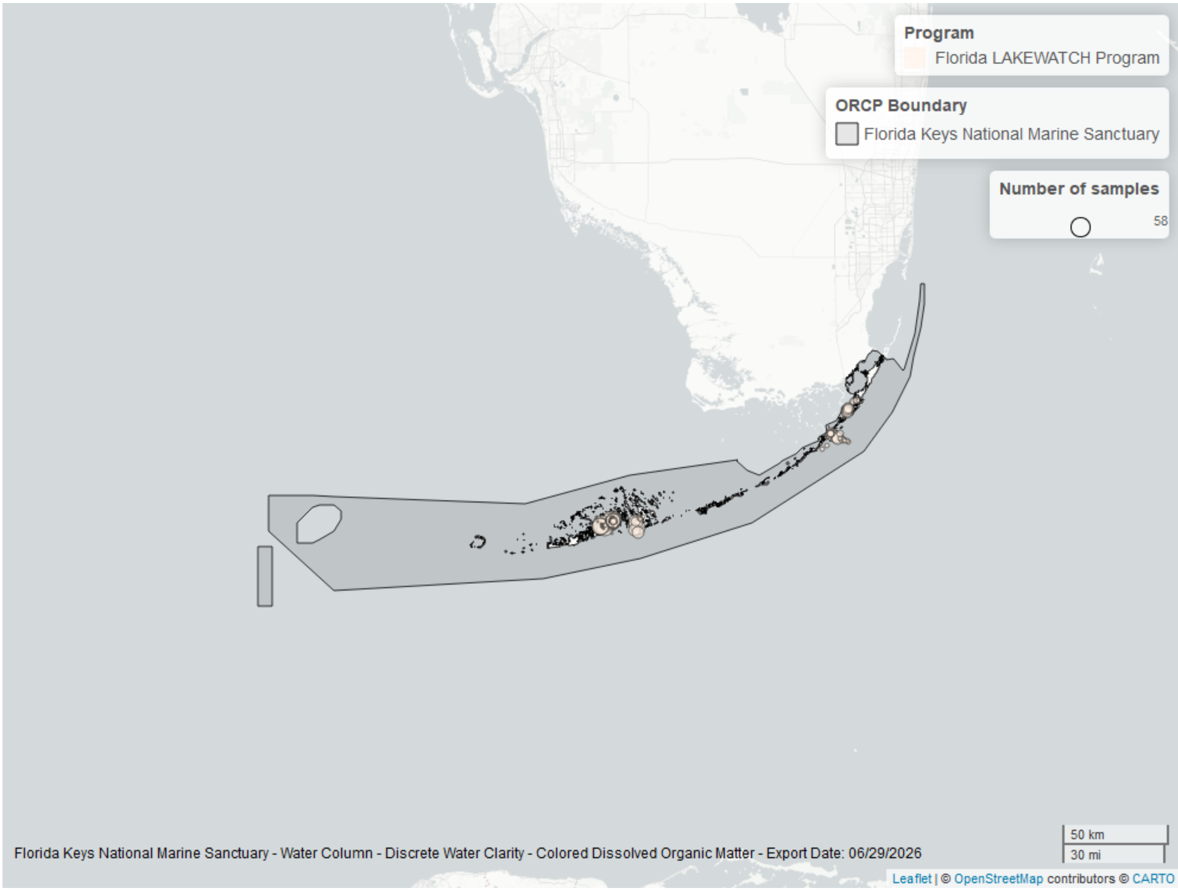

Figure 54: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.